Hold On to Your Cash:

CEO Speech Uncertainty and Financing Conservatism

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Abstract

We extend the three-component CEO speech-uncertainty decomposition of Dzielínski, Wagner, and Zeckhauser (2021)—a persistent CEO-style trait (`ClarityCEO`, the negative of a CEO fixed effect estimated from prepared-remarks segments), a call-level state residual within the Q&A segment (`UncResCEO`), and a presentation-segment uncertainty share (`UncPreCEO`)—from the market-reaction, analyst-behavior, firm-performance, and CEO-compensation outcomes on which Dzielínski et al. validate it to a class of outcomes they do not test: firm-level financing-policy decisions. Using a sample of 112,968 earnings-call transcripts covering 2,429 U.S. firms over fiscal years 2002–2018, we estimate the canonical precautionary-cash response of Opler, Pinkowitz, Stulz, and Williamson (1999) and Bates, Kahle, and Stulz (2009) on the three components jointly. The Q&A residual `UncResCEO` loads positively on cash holdings in all twelve specifications of a four-step fixed-effects panel ($p < 0.10$ one-tailed; eight of twelve at $p < 0.05$ in the Full method, twelve of twelve at $p < 0.10$ and seven of twelve at $p < 0.05$ in the QtrExp expanding-window method), and persistent CEO style loads negatively in nine of twelve specifications under both methods. The state-residual response amplifies among credit-constrained Unrated firms on the one-quarter-lead dependent variable (two of eight cells at $p < 0.05$ in the Faulkender and Petersen (2006) interaction), and among firms in the upper half of the cash-flow-volatility distribution (Han & Qiu, 2007) at one-tier-weaker significance. Construct-validation regressions show CEO speech uncertainty itself responds positively to firm-level political risk (Hassan, Hollander, van Lent, & Tahoun, 2020), US and global news-based economic-policy-uncertainty indices (Baker, Bloom, & Davis, 2016; Davis, 2016), and product-market competition (Hoberg & Phillips, 2016). Outside-world reaction regressions show that the post-call quoted bid-ask spread (4 of 12 specifications at $p < 0.10$) and SEC comment-letter receipt (4 of 6 specifications at $p < 0.10$) both load *presentation-segment* uncertainty—the segment Dzielínski et al. characterize as scripted firm-culture-reflecting—rather than the Q&A components. Three identification designs (CEO sudden-death difference-in-differences, Dzielínski et al. §6 first-difference replication around CEO turnover, and Lewbel (2012) heteroskedasticity-based instrumental-variable estimation) address reverse-causality, time-invariant heterogeneity, and time-varying omitted-variable threats respectively.

Keywords: CEO speech uncertainty, earnings calls, precautionary cash holdings, financing conservatism, credit constraints, instrumental variables, textual analysis.

JEL classification: G14, G32, G34, M41, D83.

1 Introduction

1.1 Motivation

Quarterly earnings calls offer a higher-frequency window into managerial communication than the annual 10-K disclosure cycle, and a richer one in source-identifiable speaker attribution: a single call admits separate prepared-remarks and spontaneous Q&A segments and identifies who is speaking (CEO, CFO, other manager, analyst). Dzielínski et al. (2021) show that the prepared and spontaneous segments carry systematically different uncertainty content, and that within the Q&A segment one can further isolate a persistent CEO-specific style component (`ClarityCEO`, recovered as the negative of a CEO fixed effect) from a call-level state residual (`UncResCEO`). Their three-component architecture is the starting point for this paper.

Dzielínski et al. validate their three-component decomposition on a nine-outcome set spanning market reactions (short- and long-window cumulative abnormal returns, abnormal trading volume), analyst behavior (forecast revisions, median recommendations), firm performance and valuation (Tobin's Q, return on assets), and governance (CEO compensation), as enumerated in their Appendix Table A.1. Their outcome set does not include firm-level financing-policy decisions. The natural next question—and the central question of this paper—is whether the same three-component speech-uncertainty signal carries information for the firm's own *financing* decisions: cash holdings, the precautionary liquidity buffer formalized by Opler et al. (1999) and updated for the post-2000 sample by Bates et al. (2009).

1.2 Gap

To our knowledge, no prior work has applied a CEO-specific three-component speech-uncertainty decomposition to firm-quarter cash holdings, nor jointly examined the firm's own cash-financing margin and the outside-world's reaction margin (post-call quoted spread, regulatory comment-letter receipt) of CEO speech uncertainty under a common decomposition. Dzielínski et al.'s outcome set spans market reactions, analyst behavior, firm performance and valuation, and CEO compensation; the firm-side cash response is conceptually distinct (insider financing decisions versus outsider price formation and analyst behavior) and theoretically anchored elsewhere (precautionary motive of Opler et al.). The CEO-specific aggregation we adopt is theoretically motivated by Dzielínski et al.'s scripted-versus-improvised distinction between the prepared-remarks and Q&A segments, together with the manager-fixed-effect-on-cash channel of Bertrand and Schoar (2003): both anchor the cash-policy response to the CEO speaker specifically rather than to a manager-pool aggregate.

1.3 Approach

We preserve the Dzielínski et al. decomposition. The Q&A speech-uncertainty share is recovered using the Loughran and McDonald (2011) finance-specific uncertainty word list at speaker-utterance resolution, aggregated over CEO speakers in the call's Q&A segment. We then estimate a CEO fixed-effect regression of this share on Bates-style controls and recover the negative of the CEO fixed effect as `ClarityCEO` (a CEO-level trait following Bertrand and Schoar (2003), who establish that manager fixed effects significantly predict corporate cash holdings) and the regression residual as `UncResCEO` (a call-level state residual). The presentation-segment analogue `UncPreCEO` is computed analogously on prepared remarks. We test the cash response (HC, §3.2) and its credit-constraint moderation (HFC, §3.3) on a four-step firm/industry × year/year-quarter fixed-effects ladder of 112,968 earnings-call transcripts covering 2,429 U.S. firms over fiscal years 2002–2018, sourced from S&P Capital IQ and merged with Compustat quarterly fundamentals and CRSP daily security data. Inference on the speech-uncertainty regressors is one-tailed in the theoretically predicted direction (Opler et al., 1999; Bertrand & Schoar, 2003; Almeida, Campello, & Weisbach, 2004).

1.4 Empirical Setting and Findings

Cash response. The Q&A residual `UncResCEO` is positive and statistically significant at the ten-percent threshold in all twelve specifications of suite H1 (eight of twelve at the five-percent threshold in the Full method; twelve of twelve at the ten-percent threshold and seven of twelve at the five-percent threshold in the QtrExp expanding-window method), with point estimates between +0.0016 and +0.0028 in Full and +0.0016 and +0.0046 in QtrExp. The persistent CEO-style trait `ClarityCEO` is negative and significant in nine of twelve specifications under both methods, with point estimates between −0.0170 and −0.0046 in Full and −0.0215 and −0.0024 in QtrExp; the QtrExp method strengthens the persistent-style loading to nine of twelve at the five-percent threshold and six of twelve at the one-percent threshold. The presentation-segment share `UncPreCEO` is null across the same panel. The headline pattern of this paper is that the three-component speech-uncertainty decomposition Dzielínski et al. introduce carries information for firm-level cash policy: *applied to a previously-untested outcome class, the same constructs deliver systematic and pre-registered loadings on the firm's precautionary cash buffer.*

Constraint moderation. Among the rated subsample with available Compustat credit-rating coverage (2002–2016), the `UncResCEO` × Unrated interaction is positive at the five-percent threshold in two of eight cells on the one-quarter-lead cash dependent variable, with the credit-constrained Unrated firms of Faulkender and Petersen (2006) amplifying the residual response by approximately three-fold over the rated pool (HFC, §3.3). A second constraint-channel moderator constructed on Han and Qiu's sixteen-quarter cash-flow-volatility measure produces the same lead-dependent-variable amplification pattern at one-tier-weaker statistical significance (§3.4); the two constraint operationalizations (Faulkender & Petersen, 2006; Han & Qiu, 2007) agree on the channel direction.

Driver-side construct validation. The CEO speech-uncertainty metric itself responds positively to four independently measured uncertainty drivers (§4.1): the firm-level political-risk index of Hassan et al. (2020), the US news-based economic-policy-uncertainty index of Baker et al. (2016), the global extension of Davis (2016), and product-market competition pressure measured by the text-based similarity index of Hoberg and Phillips (2016).

Outside-world reaction. The post-call closing-quote relative bid-ask spread averaged over the 25-trading-day window of Bushee, Gow, and Taylor (2018) (§4.2.1) and the within-quarter SEC comment-letter receipt indicator of Lerman, Steffen, and Zhang (2026) (§4.2.2) both load the *presentation-segment* uncertainty component `UncPreCEO`—4 of 12 spread cells and 4 of 6 SEC cells significant at $p < 0.10$ one-tailed—with the persistent style and Q&A residual components null in both outside-world regressions. The asymmetry sits cleanly on Dzielínski et al.'s own theoretical structure: the segment outsiders react to is the one Dzielínski et al. characterize as "scripted" and "firm-culture-reflecting," while the segments that drive the firm's cash decisions (Q&A persistent and Q&A residual) are the ones they characterize as "revealing CEO style."

Endogeneity. Section 4.3 reports three identification designs against three orthogonal threats. A CEO sudden-death difference-in-differences (§4.3.2) identifies a reverse-causality-resistant exogenous shock at small but cleanly-identified treated-firm count. A first-difference replication of Dzielínski et al.'s §6 specification around CEO turnover (§4.3.3) addresses time-invariant firm and manager heterogeneity. A Lewbel (2012) heteroskedasticity-based instrumental-variable estimator (§4.3.4) addresses time-varying omitted confounders. The three designs cover orthogonal identification threats; we frame this as a three-threats package rather than as convergence on a single point estimate.

1.5 Contributions

The paper makes three contributions.

1. **Extending the DWZ three-component decomposition to corporate financing-policy outcomes.** Dzielínski et al. (2021) introduce and validate the persistent-trait, Q&A-residual, presentation-share decomposition on market reactions, analyst forecasts and recommendations, firm value and performance, and CEO compensation; their paper does not test firm-level financing-policy outcomes. We apply their three-component architecture to the canonical precautionary-cash response of Opler et al. (1999) and Bates et al. (2009) and document that the Q&A residual `UncResCEO` loads positively on cash in all twelve fixed-effects specifications (§3.2). The decomposition's three components carry pre-registered, theoretically grounded signal for firm-internal financing decisions in addition to the market and governance outcomes Dzielínski et al. document.

2. **Insider-outsider channel asymmetry on a common decomposition.** The same three-component decomposition that produces the cash-side result reproduces an insider-outsider channel asymmetry on the outsider-reaction outcomes (§4.2). Insiders' cash decisions track Q&A components (`ClarityCEO` and `UncResCEO`); the post-call spread and SEC comment-letter outsiders track the presentation-segment component (`UncPreCEO`). The asymmetry is not imposed; it emerges within Dzielínski et al.'s own scripted-versus-improvised theoretical structure.

3. **Three-threats endogeneity package.** Three identification designs cover orthogonal threats to identification of the speech-uncertainty effect on cash (§4.3). The package is presented as covering three threats with three designs, not as convergence on a single point estimate.

1.6 Roadmap

Section 2 develops the conceptual framework and precommitment statement (§2.1), the speech-uncertainty measurement architecture (§2.3), the precautionary-motive and constraint-amplification literature anchors for HC and HFC (§2.4), and the empirical design (§2.5). Section 3 reports the data, sample, variables, and specification (§3.1); the cash-holdings result (HC, §3.2); the credit-constraint moderation (HFC, §3.3); and the cash-flow-volatility moderator (§3.4). Section 4 reports the four driver regressions (§4.1), the post-call spread reaction (§4.2.1), the SEC comment-letter reaction (§4.2.2), and the three-threats endogeneity package (§4.3). Section 5 synthesizes the channel asymmetries, discusses limitations, and outlines future research.

2 Conceptual Framework and Empirical Strategy

2.1 Pre-Commitment Statement

We pre-commit to the speech-uncertainty decomposition of Dzielínski et al. (2021)—a persistent CEO-style trait `ClarityCEO` (the negative of a CEO fixed effect estimated from the prepared-remarks regression), a call-level state residual `UncResCEO` from the same regression, and a presentation-segment uncertainty share `UncPreCEO`—and we test the three components on cash holdings (HC) and constraint-moderated cash holdings (HFC) jointly within each regression specification. Three opposite-sign predictions are pre-registered with differing levels of theoretical commitment.

1. **`UncResCEO` → cash↑.** Positive one-tailed; *pre-registered primary*. The directional prediction follows the precautionary chain of Opler et al. (1999), who establish that uncertainty raises optimal cash, retained for the modern post-1980 cash-secular trend by Bates et al. (2009).

2. **`ClarityCEO` → cash.** Channel-of-existence pre-registered; *specific direction interpretive*. Bertrand and Schoar (2003) establish that manager fixed effects significantly predict corporate cash holdings, anchoring the existence of a CEO-trait → cash channel without committing ex ante to a particular sign for our `ClarityCEO` construct.

3. **`UncPreCEO` → cash null.** *Null expected once Q&A components are jointly included*. Dzielínski et al. themselves treat the presentation-segment share as a control reflecting "uncertainty in communication resulting from persistent firm characteristics" rather than as a channel-of-interest in the Q&A-residual regression.

For the constraint moderator (HFC), we pre-register `UncResCEO` × Unrated > 0 (one-tailed) on the contemporaneous and one-quarter-lead cash dependent variable, with the credit-constrained Unrated firms of Faulkender and Petersen (2006) amplifying the residual response in the manner of the macro-shock asymmetry of Almeida et al. (2004) Section II.D. A secondary constraint moderator built on the cash-flow-volatility metric of Han and Qiu (2007) is pre-registered in the same direction at one tier weaker.

The three sets of cells we report in Section 3 (HC, HFC, and the cash-flow-volatility moderator) and the three identification designs we report in Section 4.3 (CEO sudden-death difference-in-differences, the Dzielínski et al. §6 first-difference replication, and the Lewbel (2012) heteroskedasticity-based instrumental-variable estimator) were fixed in advance of estimation. The endogeneity package is framed as covering three orthogonal identification threats with three pre-specified designs, not as convergence on a single point estimate.

2.2 Conceptual Framework: Precautionary Motive

Cash buffers firms against costly external financing under uncertainty. Opler et al. (1999) formalize this as the precautionary motive: "cash flow shortfalls might prevent a firm from investing in profitable projects if the firm does not have liquid assets," so firms with greater cash flow uncertainty hold more cash. Bates et al. (2009) document that the average cash-to-assets ratio of U.S. industrial firms more than doubles from 1980 to 2006, retaining the precautionary motive as the principal explanation; we adopt the same empirical lens for our 2002–2018 sample. The traditional measurement of uncertainty in this literature is historic cash-flow volatility—backward-looking and accounting-based—which we extend in Section 2.3 to a real-time text-based proxy from CEO speech during earnings calls. Minton and Wruck (2001) provide the title label "financial conservatism" and document the empirical cluster pattern (low leverage, high cash) we extend to the speech-uncertainty cause. We disclose the mechanism split: the cluster pattern is consistent with the precautionary motive of Opler et al. but also with pecking-order financial slack; we follow the precautionary mechanism throughout.

2.3 Speech-Uncertainty Measurement Architecture

The three-component speech-uncertainty decomposition follows Dzielínski et al. (2021). The Q&A speech-uncertainty share is computed at speaker-utterance resolution using the Loughran and McDonald (2011) finance-specific uncertainty word list, aggregated over CEO speakers in the call's Q&A segment. Dzielínski et al. estimate a CEO fixed-effect regression of this share on Bates-style controls and recover the negative of the CEO fixed effect as the persistent CEO-style trait `ClarityCEO`, with the same regression's residual interpreted as the call-level state-uncertainty signal `UncResCEO`. The presentation-segment analogue `UncPreCEO` is computed analogously on the prepared-remarks segment.

The two segments of the call admit theoretically distinct interpretations within Dzielínski et al.'s framework. The presentation segment is "scripted and vetted beforehand... likely to reflect the firm's culture"; the Q&A segment is "inevitably somewhat improvised and hence more likely to reveal each CEO's style." Within Q&A, `ClarityCEO` captures the time-invariant CEO trait that survives controls, while `UncResCEO` captures the time-varying call-state signal orthogonal to it. Dzielínski et al. characterize the latter as "residual uncertainty that is not explained by any of the control variables included in the regression" and document, on their stock-price-reaction outcome set, that this residual "explains little of the market reaction." Dzielínski et al. validate the three-component decomposition on market reactions, analyst behavior, firm performance, and CEO compensation; the headline empirical question of this thesis is whether the same architecture carries information for the firm's own cash-financing decision—an outcome class their paper does not address.

We preserve the CEO-only aggregation throughout the body for two theoretical reasons. First, Dzielínski et al.'s scripted-versus-improvised distinction localizes within-call CEO style specifically to the Q&A segment of the call, where the CEO speaker is the salient agent for the firm-side response. Second, Bertrand and Schoar (2003) establish a manager-fixed-effect channel on corporate cash holdings, in which the CEO trait is the load-bearing manager identity for the cash-policy outcome. Both anchors point to the CEO speaker as the theoretically appropriate aggregation level for a cash-policy outcome variable. Construct-validity evidence—that the CEO speech-uncertainty metric responds positively to firm-level political risk (Hassan et al., 2020), the U.S. news-based economic-policy-uncertainty index (Baker et al., 2016), the global extension of Davis (2016), and product-market competition pressure (Hoberg & Phillips, 2016)—is reported in Section 4.1 before the firm-side outcome tests.

2.4 Precautionary Motive and Constraint Amplification: Hypothesis Development

Hypothesis 1 (HC) — Cash response under the DWZ three-component decomposition. The precautionary chain of Opler et al. and Bates et al. delivers `UncResCEO` $\rightarrow$ cash$\uparrow$ as the primary directional prediction: uncertainty raises optimal cash, the Q&A residual is a real-time uncertainty signal at the call-level resolution, and within-call CEO uncertainty therefore should load on the firm's precautionary cash buffer. The persistent CEO-style trait `ClarityCEO` is anchored on the existence-proof channel of Bertrand and Schoar (2003), who document that manager fixed effects significantly predict corporate cash holdings (the adjusted R-squared of the cash-holdings regression rises from 77% to 80% on inclusion of manager fixed effects, with F-test rejection of the all-zero null at $p \leq .0001$ for CEOs, CFOs, and other executives), with documented systematic heterogeneity that "some managers prefer holding relatively less debt and more cash than other managers." We pre-register the existence of the CEO-trait $\rightarrow$ cash channel; the specific direction we observe is reported but not pre-registered as a directional theoretical prediction. Dzielínski et al. treat `UncPreCEO` as a control variable in the Q&A-residual regression—reflecting "uncertainty in communication resulting from persistent firm characteristics"—and we accordingly expect a null on the presentation-segment uncertainty share once the Q&A components are jointly included.

Hypothesis 2 (HFC) — Constraint amplification of `UncResCEO` on cash. Almeida et al. (2004) test in their Section II.D ("Dynamics of Liquidity Management: Responses to Macroeconomic Shocks", p. 1800) the asymmetry that "constrained firms save significantly larger fractions of their cash flows than unconstrained firms do following those shocks", a result they restate in their concluding remarks (p. 1802) as: "financially constrained firms should increase their propensity to retain cash following negative macroeconomic shocks, while unconstrained firms should not." We adopt this asymmetry as the H2 anchor, with within-call CEO Q&A residual uncertainty playing the role of the "shock" trigger: firms unable to access external capital must self-finance through retained cash buffers, amplifying the contemporaneous and lead-quarter cash response to a within-call uncertainty signal. Faulkender and Petersen (2006) provide the operational definition we use: firms without a credit rating are credit constrained, exhibiting significantly less debt and slightly more equity than rated firms (Faulkender–Petersen verbatim: "credit constrained," not "capital constrained"). We adopt their binary rated-versus-unrated scheme directly: the Unrated indicator equals one for firms without a credit rating, and the rated reference category pools investment-grade and below-investment-grade firms.

A secondary moderator built on Han and Qiu (2007) tests the same constraint channel through a different proxy: firms in the upper half of the cash-flow-volatility distribution face stronger financial-constraint shocks at the firm-quarter resolution than rating provides at the firm level. The Q&A residual signal should amplify on the constrained (high-volatility) tail in the same direction as the rating moderator, at one-tier-weaker statistical significance.

Honest disclosures. The `ClarityCEO` direction observed in Section 3.2 is interpretive rather than pre-registered. The existence of the CEO-trait → cash channel is anchored on Bertrand and Schoar (2003); the specific negative sign we observe is reported as an empirical pattern interpretable through Dzielínski et al.'s framing of `ClarityCEO` as the stable trait "not motivated by business uncertainty," but neither anchor predicts the sign ex ante. The `UncPreCEO` → cash null we predict is a low-confidence claim relying on the control-variable role Dzielínski et al. assign to the presentation segment. The H2 amplification anchor is Almeida et al.'s *macro*-shock asymmetry; our extension to within-call CEO speech uncertainty as the shock signal is a hypothesis-development extension, not a pre-existing theoretical result. Identification of the speech-uncertainty effect on cash is addressed in Section 4.3 via three orthogonal designs covering reverse-causality, time-invariant heterogeneity, and time-varying omitted-confounder threats; we do not claim convergence on a single point estimate from the package.

2.5 Empirical Design

We estimate a four-step fixed-effects ladder of panel-OLS regressions of cash-to-assets on the three speech-uncertainty components and Bates-style controls. The four FE configurations cross firm or industry (Fama-French 12) entity effects with calendar-year or calendar year-quarter time effects. Standard errors are clustered at the firm level for the firm-side regressions; the macro-driver suites of Section 4.1 use two-way clustering on firm and calendar quarter to handle within-quarter cross-firm correlation in the macro regressors. The base control vector follows Bates et al. (size, leverage, return on assets, dividend dummy, cash-flow volatility, capital expenditure, R&D, working capital, market-to-book); the extended control vector adds the constructs reported in Appendix A. Both contemporaneous and one-quarter-lead cash dependent variables are reported throughout. Inference on each speech-uncertainty regressor is one-tailed in the theoretically predicted direction (Opler et al., 1999; Bertrand & Schoar, 2003; Almeida et al., 2004), with constraint-moderator interactions tested one-tailed positive per the Almeida et al. amplification asymmetry. Formal hypothesis statements (HC, HFC, and H1.3 CFvol) and the four-step FE ladder are summarized in Section 3.1.

3 Main Empirical Analyses

3.1 Data, Sample, Variables, Specification

Sample. The estimation sample is a firm-quarter panel of 112,968 earnings-call transcripts covering 2,429 U.S. firms over fiscal years 2002–2018. Earnings-call transcripts are sourced from S&P Capital IQ; Compustat quarterly fundamentals supply the financial covariates and CRSP daily security data the market controls. Financial firms (SIC 6000–6999) and regulated utilities (SIC 4900–4999, retaining 4953 and 4959) are excluded per Bates et al.'s sample-construction conventions. Firm-quarters with missing values on the dependent variable, on any base control, or on the speech-uncertainty constructs are dropped listwise.

Dependent variable. Cash-to-assets is computed as Compustat `cheq / atq` at quarterly frequency, the operational form of Bates et al. (2009), who explicitly evaluate and reject Opler et al.'s log-of-cash-to-net-assets in their 1980–2006 sample. Both the contemporaneous cash ratio (`CashRatio`) and its one-quarter lead (`CashRatio_lead`) are reported. The lead-DV specifications are the more demanding test of a directional uncertainty → cash-buildup channel: a positive coefficient on a lead-DV specification reflects a contemporaneous-quarter signal predicting next-quarter cash, with the lagged dependent variable absorbing within-firm cash-stock persistence.

Speech-uncertainty independent variables. The three components of the Dzielínski et al. (2021) decomposition (`ClarityCEO`, `UncResCEO`, `UncPreCEO`) are constructed as detailed in Section 2.3 and Appendix A. Each component is standardized (mean zero, unit variance) within the estimation sample so that the reported β coefficients have the interpretation "effect of a one-standard-deviation change in the speech-uncertainty component on the firm's cash ratio." The three components enter each specification jointly. We report two parallel measurement methods. The *Full* method estimates the CEO fixed-effect regression on the full firm-quarter panel; the *QtrExp* method estimates an expanding-window CEO fixed effect on within-CEO data only, using only the calls of a given CEO available up to and including the focal call. The QtrExp method is strictly look-ahead-free and is the secondary-identification cross-check for the persistent style component `ClarityCEO`.

Controls. The base control vector follows Bates et al. (2009): lnAssets, market-to-book, return-on-assets, dividend dummy, leverage, daily stock volatility, capital expenditure-to-assets, R&D-to-sales, and net-working-capital-to-assets, all measured at the call's fiscal-quarter end. The extended control vector adds eight additional constructs (acquisition activity, equity issuance flag, share repurchase, accruals-quality, post-call bid-ask spread, analyst dispersion, earnings-surprise magnitude, and segment count); see Appendix A for full operational definitions. The lagged dependent variable is included as a base control in all specifications.

Specification ladder. Each suite reports a four-step fixed-effects ladder of panel-OLS regressions. The four FE configurations cross firm or industry (Fama-French 12) entity effects with calendar year or calendar year-quarter time effects; for each entity-time crossing, both the contemporaneous and the one-quarter-lead cash dependent variable are reported, yielding twelve cells per suite (HC) or eight cells (HFC and CFvol, where the constraint moderator interacts with two of the three speech-uncertainty components). Standard errors are clustered at the firm level. Inference on each speech-uncertainty regressor is one-tailed in the theoretically predicted direction (`UncResCEO` $>$ 0 per the precautionary chain; `UncResCEO` $\times$ Unrated $>$ 0 per the constraint asymmetry); the persistent style trait `ClarityCEO` and the presentation share `UncPreCEO` are reported one-tailed in their observed-empirical direction with the directional-versus-existence distinction documented in Section 2.1.

3.2 Cash Holdings under the DWZ Three-Component Decomposition

Table 8 (Full method) and Table 9 (QtrExp method) report the four-step fixed-effects ladder of cash-ratio regressions on the three CEO speech-uncertainty components. Dzieliński et al. establish their decomposition on market and governance outcomes; the present analysis extends the architecture to a domain their paper does not test, the firm's own cash-financing decision:

> The Q&A residual `UncResCEO` loads positively in 12 of 12 specifications at $p < 0.10$ one-tailed (8 of 12 at $p < 0.05$ in the Full method; 12 of 12 at $p < 0.10$ and 7 of 12 at $p < 0.05$ in the QtrExp method), with point estimates between +0.0016 and +0.0028 in Full and +0.0016 and +0.0046 in QtrExp. The Q&A residual Dzieliński et al. introduce as a call-level state-uncertainty signal carries pre-registered, theoretically grounded signal for the firm's precautionary cash buffer.

The persistent CEO-style trait `ClarityCEO` loads in the opposite direction. In the Full method (Table 8), `ClarityCEO` is negative and significant in 9 of 12 specifications at $p < 0.10$ (5 of 12 at $p < 0.05$), with point estimates between −0.0170 and −0.0046. In the QtrExp method (Table 9), the persistent-style channel sharpens: `ClarityCEO_QtrExp` is negative and significant in 9 of 12 specifications at $p < 0.10$, 9 of 12 at $p < 0.05$, and 6 of 12 at $p < 0.01$, with point estimates between −0.0215 and −0.0024. The empirical interpretation under the Bertrand and Schoar channel-of-existence anchor is that CEOs whose persistent communication style is more uncertain (lower Clarity) preside over firms with higher cash holdings; the specific negative sign is reported as observed and not pre-registered as a directional theoretical prediction (Section 2.4 disclosure).

The presentation-segment uncertainty share `UncPreCEO` is null on cash in both methods (0 of 12 specifications at any threshold, with point estimates spanning a tight range of −0.0007 to +0.0009 in Full and −0.0004 to +0.0016 in QtrExp). Per Dzieliński et al., the prepared-remarks share controls for "uncertainty in communication resulting from persistent firm characteristics"; once the Q&A components are jointly included, the presentation share carries no marginal information for the cash-financing decision.

The full-versus-QtrExp comparison is directly informative on the within-CEO identifying variation behind `ClarityCEO`. The Full method estimates the CEO fixed effect on the entire firm-quarter panel (the original Dzieliński et al. construction); the Qtr-

Exp method estimates it on the within-CEO panel up to the focal call, removing any look-ahead. The two methods agree in direction on all three components and yield slightly stronger statistical evidence for `ClarityCEO` in the QtrExp method (which uses fewer observations and a more demanding within-CEO identification window). The agreement supports interpreting `ClarityCEO` as a CEO-trait signal rather than a panel-fitted artifact.

3.3 Constraint Amplification: Unrated Channel

Table 10 (Full method) and Table 11 (QtrExp method) report the eight-cell HFC ladder, which interacts the credit-constraint moderator (the Unrated indicator of Faulkender and Petersen (2006)) with each speech-uncertainty component. The sample is restricted to firms with available Compustat credit-rating coverage (2002–2016), yielding panel sizes between 38,092 and 38,737 in Full and between 21,371 and 21,824 in QtrExp.

The base `UncResCEO_c` effect on cash within the rated subsample is positive and significant in 8 of 8 specifications at $p < 0.10$ in Full (6 of 8 at $p < 0.05$) and 8 of 8 at $p < 0.10$ in QtrExp (6 of 8 at $p < 0.05$), with point estimates between +0.0018 and +0.0027 in Full and +0.0021 and +0.0044 in QtrExp. The base effect on the constraint-restricted subsample replicates the HC main result.

The interaction `UncResCEO_c` × Unrated is positive and significant in 2 of 8 specifications at $p < 0.05$ in the Full method (cells 5 and 7, both lead-DV specifications), with point estimates ranging from −0.0001 to +0.0057. The QtrExp method delivers a parallel pattern with one-tier-weaker statistical evidence: 2 of 8 cells at $p < 0.10$ on the lead-DV specifications, 0 of 8 at $p < 0.05$, with point estimates from +0.0013 to +0.0050. The contemporaneous-DV interaction terms are not statistically distinguishable from zero in either method.

The empirical pattern is the asymmetric amplification predicted by the H2 anchor of Almeida et al. (2004), transposed from the macro-shock setting to a within-call CEO speech-uncertainty signal. Credit-constrained Unrated firms exhibit a larger one-quarter-lead cash-buildup response to a within-call uncertainty signal than rated firms do; the contemporaneous response is not differentially amplified, consistent with the constraint channel operating through next-quarter financing-need realization rather than within-quarter cash repositioning. The presentation-segment interaction `UncPreCEO_c` × Unrated is significant in only 1 of 8 cells at $p < 0.10$ in Full and 0 of 8 in QtrExp; the constraint moderator does not activate the presentation share. The persistent-style × Unrated interaction is not reported as a primary cell because the trait-versus-trait product (Clarity is a CEO trait; Unrated is a firm-credit trait) does not have a direct theoretical mapping to the macro-shock asymmetry literature.

3.4 Constraint Amplification: Cash-Flow-Volatility Channel

Table 12 reports the eight-cell ladder built on the cash-flow-volatility moderator of Han and Qiu (2007), an alternative operational definition of the financial-constraint channel. The HighCFvol indicator is the upper-half of the firm-quarter cash-flow-volatility distribution (sixteen-quarter rolling coefficient of variation); the constraint channel should activate on the same direction as the rating moderator at the firm-quarter resolution rating cannot reach.

The base `UncResCEO_c` effect is positive and significant in 6 of 8 specifications at $p < 0.10$ (4 of 8 at $p < 0.05$), with point estimates between +0.0015 and +0.0025. The persistent style `ClarityCEO_c` loads negatively in 6 of 8 cells at $p < 0.10$ (3 of 8 at $p < 0.05$, 1 of 8 at $p < 0.01$), with point estimates between −0.0135 and −0.0058. The presentation share `UncPreCEO_c` is null on the base effect (0 of 8 at any threshold).

The pre-registered moderator interaction `UncResCEO_c` × HighCFvol is positive in 2 of 8 cells at $p < 0.10$ (the lead-DV industry-FE specifications), 0 of 8 at $p < 0.05$, with point estimates from −0.0000 to +0.0048. The pattern matches the rating moderator at one-tier-weaker significance, consistent with the H2 prediction that the cash-flow-volatility proxy captures the same constraint channel through a different identifying margin.

A second interaction surfaces a finding not pre-registered. The presentation-segment uncertainty share interacted with HighCFvol, `UncPreCEO_c` × HighCFvol, is positive and significant in 4 of 8 cells at $p < 0.10$, 4 of 8 at $p < 0.05$, and 2 of 8 at $p < 0.01$, with point estimates between +0.0002 and +0.0101. `UncPreCEO_c` is null on the base effect, but its interaction with the cash-flow-volatility tail activates a presentation-segment cash-amplification channel. The interaction sign is in the same direction as the residual-component interaction; at the upper end of the point-estimate range the presentation-segment interaction is roughly double the residual-component interaction (+0.0101 versus +0.0048). The interpretation is that high-cash-flow-volatility firms read the prepared-remarks uncertainty share as additional state-level signal precisely when their constraint exposure is highest, even though the base prepared-remarks share is null. We treat this finding as robustness-extending, not channel-redirecting: Dzielínski et al.'s decomposition treats the presentation share as a control in the Q&A-residual regression but does not preclude an interaction-context activation in a constraint-moderated cash setting. The pattern is internally consistent with the same asymmetry our HC and HFC results document, with the prepared-remarks share contributing in the constraint-tail rather than at the mean.

3.5 Robustness Notes

Two robustness disclosures are documented. First, the four-step fixed-effects ladder is itself the within-regressor robustness frame: every column of every body table reports a different point on the (firm or industry) × (year or year-quarter) cross, so the FE-ladder check is integrated into the main display rather than a separate appendix specification (Appendix B.3). Second, the linear cash-to-assets dependent variable of Bates et al. is the body specification; an OPSW log-of-cash-to-net-assets robustness is flagged as a planned extension in Section 5 but is not estimated in the present draft (Appendix B.2).

4 Additional Analyses

4.1 Drivers of CEO Speech Uncertainty

The cash-financing and outside-world results in Sections 3 and 4.2 treat the speech-uncertainty constructs as right-hand-side regressors. Their interpretation as legitimate uncertainty proxies depends on whether they respond, on the left-hand side, to independently measured exogenous uncertainty drivers. We test four such drivers as construct-validation regressions of the speech-uncertainty constructs on (i) the firm-level political-risk index of Hassan et al. (2020), (ii) the U.S. news-based economic-policy-uncertainty index of Baker et al. (2016), (iii) the global extension of Davis (2016), and (iv) product-market competition pressure as measured by the text-based total similarity index of Hoberg and Phillips (2016). Each driver regression is reported across four cells: the two CEO speech-uncertainty components from the Dzielínski et al. decomposition (UncResCEO Q&A residual and UncPreCEO presentation segment) crossed with two entity fixed-effect configurations (industry vs. firm). The macro-driver suites (BBD-EPU and GEPU) use two-way clustering on firm and calendar quarter to handle within-quarter cross-firm correlation in the macro regressor, per Section 2.5. The four drivers test construct validity rather than formal hypotheses.

Political risk (Hassan et al., 2020). The contemporaneous firm-level political-risk index loads positively on both CEO speech-uncertainty components in all four cells of Table 13 at $p < 0.01$ one-tailed, with point estimates between +0.0001 and +0.0003. The two-quarter-lag specification (Table 14) is positive in all four cells, with the two UncPreCEO cells significant at $p < 0.01$ ($\beta \approx +1.5 \times 10^{-4}$ industry FE, $\beta \approx +3.7 \times 10^{-5}$ firm FE); the two UncResCEO cells are positive but non-significant at $p < 0.10$ (point estimates near zero, $p_{\text{one}} \approx 0.18$–$0.20$). The lag specification establishes that the firm-level political-risk signal is a leading driver of the presentation-segment uncertainty share, not merely a contemporaneous correlation; the Q&A-residual share carries no two-quarter-ahead political-risk signal.

Macro economic-policy uncertainty (Baker, Bloom, and Davis, 2016; Davis, 2016). The U.S. economic-policy-uncertainty index (Table 16) loads positively on both CEO speech-uncertainty components in all four cells, with all four significant at $p < 0.10$ (three of four at $p < 0.05$ and two of four at $p < 0.01$), point estimates between +0.0123 and +0.0239. The global extension (Table 17) is similarly positive in all four cells, with all four significant at $p < 0.05$ (two of four at $p < 0.01$), point estimates between +0.0181 and +0.0309. The two macro indices both deliver the predicted construct-validation pattern: the CEO speech-uncertainty components co-vary with independently measured macro uncertainty.

Product-market competition (Hoberg and Phillips, 2016). The text-based total-similarity index loads positively in all four cells of Table 15 but is concentrated in the presentation-segment component: both UncPreCEO cells are significant at $p < 0.01$ ($\beta = +0.0304$ industry FE, $\beta = +0.0302$ firm FE), while the two UncResCEO cells are positive but non-significant at $p < 0.10$ ($\beta = +0.0008$ industry FE with $p_{\text{one}} = 0.29$, $\beta = +0.0023$ firm FE with $p_{\text{one}} = 0.38$). The Pres-channel concentration is consistent with Dzielínski et al.'s framing of the prepared-remarks segment as the firm-culture-reflecting share: competition pressure manifests in the segment of the call constructed in advance with the firm's broader competitive context in mind, while the Q&A residual carries no detectable competition signal.

Synthesis. All four uncertainty drivers load positively in every cell, with $p < 0.10$ significance covering 4 of 4 (political risk), 4 of 4 (US-EPU), 4 of 4 (GEPU), and 2 of 4 (competition) cells. The two non-significant competition cells are both UncResCEO cells ($p_{\text{one}} > 0.28$); none of the sixteen driver–component cells reverses sign. The political-risk lag specification confirms timing for the presentation-segment component. Sign and significance evidence across the four drivers establishes the two CEO speech-uncertainty components as legitimate uncertainty proxies before the cash-financing and outside-world outcome tests in Sections 3 and 4.2.

4.2 Outsider Reaction: Spread and SEC Comment Letter

Section 3 reports the firm-side cash-financing response to CEO speech uncertainty under the three-component decomposition. We now turn to two outsider-reaction outcomes that Dzielínski et al. do not test, both at the firm-quarter resolution: the post-call quoted bid-ask spread (a market-microstructure outcome attributable to outside traders' information-processing costs) and the receipt of an SEC comment letter referencing the earnings call (a regulatory outcome attributable to SEC reviewers' attention to call content). The two outsider-reaction tests adopt the same three-component decomposition as the firm-side cash test, jointly entered as right-hand-side regressors with Bates-style controls. The two tests extend the DWZ measurement architecture to two new outcome classes: market microstructure beyond the abnormal-volume metric Dzielínski et al. use, and regulatory-reaction outcomes their paper does not address.

4.2.1 Market Channel: Post-Call Bid-Ask Spread

The post-call quoted bid-ask spread aggregates outside traders' information-asymmetry response to call content over the post-disclosure information-incorporation window. Bushee et al. (2018) construct an information-asymmetry measure as the average value of the Amihud (2002) illiquidity construct "over the period starting the day of the call and ending 25 trading days subsequent to the call." Because Bushee et al.'s alternative spread-side measure uses an intraday TAQ-based structural estimator that is not feasible in our daily-CRSP setting, we adapt their post-call 25-trading-day window definition (day 0 inclusive, per Bushee et al. verbatim) to a daily closing-quote relative bid-ask spread averaged over the window. Both the Bushee et al. window and the daily-CRSP-feasible spread formula are documented in Appendix A.

Table 18 reports the four-step fixed-effects ladder of post-call-spread regressions on the three speech-uncertainty components. The presentation share `UncPreCEO` loads positively in 4 of 12 specifications at $p < 0.10$, with 3 of 12 at $p < 0.05$ and 2 of 12 at $p < 0.01$. Per-cell inspection of the suite specification confirms the four significant cells are uniformly positive: cols 1 ($\beta = +0.1664$, $p_{\text{one}} = 0.0393$), 2 ($\beta = +0.3496$, $p_{\text{one}} = 0.0026$), 4 ($\beta = +0.2945$, $p_{\text{one}} = 0.0083$), and 6 ($\beta = +0.1644$, $p_{\text{one}} = 0.0825$), all on the contemporaneous post-call-spread dependent variable. The eight cells not statistically distinguishable from zero include four lead-DV cells with negative point estimates, all of them non-significant at $p < 0.10$. The presentation-segment uncertainty share predicts the call-quarter post-call spread but does not predict the next-quarter spread; the effect is contemporaneous-only, consistent with information-processing costs absorbed within the BGT post-disclosure window.

The Q&A components are null. `ClarityCEO` is not statistically distinguishable from zero in 0 of 12 cells at $p < 0.10$, with point estimates spanning -0.1886 to $+1.1056$ across cells (both signs across cells, all non-significant). `UncResCEO` is null in 0 of 12 cells at $p < 0.10$, with point estimates between -0.1021 and $+0.0984$; the contemporaneous cells uniformly negative (-0.10 to -0.06) and the lead cells uniformly positive ($+0.05$ to $+0.10$), but neither margin distinguishable from zero.

4.2.2 Regulatory Channel: SEC Comment Letter Receipt

Following Lerman et al. (2026), we measure SEC scrutiny attributable to call content using the conference-call-comment-letter (CCCL) indicator: an indicator that the firm receives an SEC comment letter referencing the focal earnings call within the call's fiscal quarter. The indicator is rare in the panel (call-quarter receipt rate well below one percent), and we estimate it as a linear-probability model in PanelOLS to maintain comparability with the rest of the body display.

Table 19 reports six specifications crossing two entity fixed-effect configurations (industry vs. firm) with two control specifications (base vs. extended) and two time-fixed-effect configurations (calendar year vs. calendar year-quarter). `UncPreCEO` loads positively on CCCL receipt in 4 of 6 cells at $p < 0.10$ (cols 2, 3, 4, 6), with 1 of 6 at $p < 0.05$ (col 3, $\beta = +0.0016$, $p_{\text{one}} = 0.0144$). Per-cell inspection of the suite specification confirms all six cells are positive (point estimates between +0.0007 and +0.0016). Three of the four extended-controls specifications (cols 3, 4, 6) survive at $p < 0.10$; col 5 (industry-FE, calendar year-quarter time effects) is the lone non-significant extended-controls cell at $p_{\text{one}} = 0.1652$. The presentation-segment loading on regulatory-reaction outcomes survives the more demanding within-firm-and-within-quarter identifying variation in three of four extended-control configurations.

The Q&A components are null on the regulatory-channel outcome, mirroring the spread-channel pattern. `ClarityCEO` is not statistically distinguishable from zero in 0 of 6 cells at $p < 0.10$, with point estimates between +0.0039 and +0.0062 but all $p_{\text{one}} > 0.50$. `UncResCEO` is null in 0 of 6 cells at $p < 0.10$, with near-zero point estimates between −0.0004 and −0.0003. Neither Q&A component carries marginal information for SEC comment-letter receipt once the presentation share is jointly included.

4.2.3 Insider-Outsider Channel Asymmetry

The Section 3 firm-side cash response loads on the Q&A components (`ClarityCEO` negative in 9 of 12 specifications, `UncResCEO` positive in 12 of 12); the Sections 4.2.1 and 4.2.2 outsider-reaction outcomes load on the presentation-segment component (`UncPreCEO` positive in 4 of 12 spread cells and 4 of 6 SEC cells, with the Q&A components null in both regressions). The asymmetry sits cleanly on Dzielínski et al.'s own theoretical structure for the two segments of the call. The presentation segment is the one Dzielínski et al. characterize as "scripted and vetted beforehand. . . likely to reflect the firm's culture"; the Q&A segment is the one they characterize as "inevitably somewhat improvised and hence more likely to reveal each CEO's style." Two segments of the same call carry signals to actors with different information sets and different decision horizons. The thesis documents this insider-outsider channel asymmetry within the DWZ framework while extending the architecture's outcome set from market reactions, analyst behavior, performance, and governance to firm financing-policy and regulatory-reaction outcomes.

4.3 Endogeneity

4.3.1 Three Designs Targeting Three Orthogonal Threats

The cash response of Section 3 faces three identification threats. First, reverse causality: cash policy could shape CEO speech rather than the reverse, with cash-rich firms speaking with calmer or clearer cadence regardless of underlying uncertainty. Second, time-invariant heterogeneity: a CEO-firm match could embed both speech style and cash policy through the matching process, with neither causing the other within match. Third, time-varying omitted confounders: a within-CEO time-varying state variable (a strategy review, a litigation flare-up, a customer-concentration shift) could move both speech uncertainty and cash policy contemporaneously, producing the within-firm correlation Section 3 documents.

We address these three threats with three designs that target them orthogonally rather than converge on a single point estimate. Subsection 4.3.2 reports a CEO sudden-death difference-in-differences against the reverse-causality threat. Subsection 4.3.3 reports a first-difference replication of Dzielínski et al.'s Section 6 turnover specification against the time-invariant heterogeneity threat. Subsection 4.3.4 reports a Lewbel (2012) heteroskedasticity-based IV against the time-varying omitted-confounders threat. Subsection 4.3.5 discloses a structural asymmetry between the ClarityCEO and UncResCEO identification problems and disposes of the asymmetric coverage of the three designs.

4.3.2 CEO Sudden-Death Difference-in-Differences

Following Bennedsen, Pérez-González, and Wolfenzon (2020), who identify CEO causal effects using exogenous shocks to CEO availability (hospitalization and death), and Ghafoor, Yousaf, and Li (2023), who apply a sudden-death DiD to firm cash holdings, we construct an exogenous-shock test on our F1D panel. We classify CEO-death events using Ghafoor et al.'s operational definition of sudden death: heart attack, plane or automobile accident, or news language indicating the death was "due to sudden illness" or "passed away unexpectedly," excluding deaths preceded by visible decline. Of the candidate events identified in the 2002–2018 panel, eight have the call-coverage and pre-event panel availability required for a ±12-quarter DiD design with at least five pre-event Q&A-segment calls per treated CEO.

Treated firms are matched 1:1 to control firms via nearest-neighbor propensity-score matching on Bates et al. (2009) covariates at $t-1$ (lnAssets, leverage, ROA, dividend dummy, scaled cash flow), drawn from the non-treated F1D pool within the same death-quarter ±2 window. Post-matching standardized differences are within the Rosenbaum–Rubin threshold of 0.25 on all five covariates. Table 20 reports four DiD specifications crossing two entity fixed-effect configurations (industry vs. firm) with two time fixed-effect configurations (none vs. year-quarter). The pooled-ATT estimate (`Treated` × `Post`) is negative across all four specifications, ranging from $\beta = -0.0053$ (firm+year-quarter FE) to $\beta = -0.0178$ (industry FE only, replicating Ghafoor et al.'s baseline specification), with one-tailed p-values between 0.121 and 0.338 on a treated-firm count of eight. The negative sign matches Ghafoor et al.'s reduced-form magnitude and the precautionary-cash prior; the magnitudes are 12–40% of Ghafoor et al.'s primary point estimate of −0.043. None of the four specifications is statistically significant at conventional one-tailed thresholds.

Three honest disclosures qualify the design at this sample size. *Pre-trend pattern.* A pre-period placebo test on the industry-FE specification estimates Treated × Pre_{4q} at $\beta = +0.027$ ($p_{\text{two}} = 0.183$) and Treated × Pre_{8q} at $\beta = +0.021$ ($p_{\text{two}} = 0.146$); both are statistically insignificant but positive in sign while the post-event ATT is negative. The pattern is consistent with mean reversion rather than a clean parallel-trends pass; we describe it as such rather than claim "parallel trends pass." *Lagged-DV exclusion.* The Phase E specifications omit the lagged dependent variable that is otherwise required in Section 3, following the standard DiD practice of Bertrand, Duflo, and Mullainathan (2004): cash is highly persistent, and including a lagged dependent variable in the DiD specification mechanically absorbs the ATT through cash-stock persistence. The firm fixed effects in cols 2 and 4 already address time-invariant heterogeneity. *Cluster count.* With sixteen unique firm clusters (eight treated plus eight matched controls), cluster-robust standard errors are below the conventional asymptotic-validity threshold of approximately thirty; we report the conventional cluster-robust standard errors with this disclosure. The originally proposed heterogeneity test on pre-event UncResCEO is not feasible at this sample size (a median split would yield four events per cell); we report the pooled ATT only and treat the speech-channel-specific decomposition as future work conditional on a larger event sample.

4.3.3 First-Difference Replication of Dzielinski, Wagner, and Zeckhauser Section 6

Dzielínski et al. address time-invariant firm and manager heterogeneity in their Section 6 by estimating a first-difference specification around CEO-turnover events: the change in the firm-level outcome is regressed on the change in average ClarityCEO between the outgoing-CEO tenure and the incoming-CEO tenure, with sample filtered to turnovers where (i) the gap between the outgoing CEO's last call and the incoming CEO's first call is at most 120 days, (ii) both CEOs have at least five calls at the firm, and (iii) both appear in the underlying CEO fixed-effect estimation window. The first-difference annihilates all firm-level and manager-level fixed components; identification is from within-turnover-pair change.

We adapt the Dzielínski et al. Section 6 design from their Tobin's Q outcome to cash holdings. The sample-construction filters yield 659 turnover pairs in the F1D panel. Table 21 reports three specifications: the verbatim Dzielínski et al. controls (col 1), the verbatim controls plus the additional Bates extras (col 2), and an industry-adjusted variant (col 3). The coefficient on ΔClarityCEO is $\beta = -0.0183$ ($p_{\text{one}} = 0.046$) in col 1, −0.0172 ($p_{\text{one}} = 0.054$) in col 2, and −0.0182 ($p_{\text{one}} = 0.046$) in col 3, each negative and significant at $p < 0.10$ one-tailed in the predicted direction (clarity rises after turnover ⇒ cash falls, the within-firm-and-within-pair-of-CEOs analogue of the within-firm pattern in Section 3.2). The Phase E and DWZ-first-difference designs converge on identical $|\beta| \approx 0.018$ across an exogenous-low-power and a weak-identification-high-power contrast on the same outcome (cash holdings) and the same speech-component (ClarityCEO).

The first-difference does not identify the UncResCEO effect because the within-CEO mean of UncResCEO is zero by construction (the residual from a CEO-fixed-effect regression has within-CEO mean exactly zero by the OLS first-order conditions; verified empirically at machine zero on our panel). The change in the per-CEO mean of UncResCEO between two CEOs is therefore identically zero in expectation, and the first-difference specification identifies ClarityCEO only. Three deviations from the Dzielínski et al. Section 6 specification are documented in Appendix A: (i) industry fixed effects use the Fama-French 12 partition rather than Dzielínski et al.'s Fama-French 48 (panel-infrastructure constraint); (ii) the intangibles-ratio control is omitted because it is not constructed in our panel; (iii) the sample restricts to F1D Main industries (Fama-French 12 excluding Finance and Utilities) per the rest of the body. Cash is also a CEO-discretion choice variable, which makes endogeneity more concerning here than in Dzielínski et al.'s Tobin's Q application. We treat this design as a robustness companion to the Phase E sudden-death DiD rather than as primary identification.

4.3.4 Heteroskedasticity-Based Instrumental Variable Estimation

The Lewbel (2012) estimator constructs instruments from heteroskedasticity in the residual of the endogenous regressor relative to a vector of base controls. Under the exclusion restriction that each base control has covariance with the squared residual of the endogenous regressor but is uncorrelated with the structural-equation error term, the heteroskedasticity-based instruments identify the structural coefficient on the endogenous regressor. The base controls in our application are the eight Bates et al. controls; we apply a heteroskedasticity-screening filter to retain only those controls whose squared residual has a statistically significant relation to the endogenous regressor's residual. Six of the eight Bates et al. controls are retained (lnAssets, leverage, Tobin's Q, ROA, capex, and the lagged cash ratio); dividend dummy and scaled cash flow are dropped at the screening stage.

Table 22 reports three specifications. Col 1 estimates the OLS baseline as a replication check: the coefficient on UncResCEO is $\beta = +0.0021$ ($p_{\text{one}} = 0.020$, $n = 43{,}471$), which replicates the cell 1 estimate of the main panel in Section 3.2 (the small observation difference relative to the main panel's $n = 43{,}333$ reflects control-NaN listwise drops in the IV-instrument residualization step). Col 2 reports the 2SLS-Lewbel estimate with the six-instrument heteroskedasticity-screened set: $\beta = +0.0100$ ($p_{\text{one}} = 0.115$, $n = 43{,}471$). Col 3 reports the 2SLS-Lewbel estimate with the extended nine-instrument set: $\beta = +0.0101$ ($p_{\text{one}} = 0.116$, $n = 43{,}454$).

The diagnostic battery is reported in the table footnote. The first-stage Cragg-Donald F-statistic is 20.4 in col 2 and 41.3 in col 3, both above the standard weak-IV threshold of $F = 10$; the col 2 statistic is below the more demanding Stock-Yogo 10%-maximal-IV-size threshold of approximately 23 for one endogenous regressor with six instruments, so we disclose the col 2 first-stage as borderline weak under the Stock-Yogo criterion. The Sargan over-identification test fails to reject the joint-validity null in col 2 ($p = 0.92$) but rejects it at $p < 0.05$ in col 3 ($p = 0.009$); we accordingly treat col 2 as the primary specification and disclose that the extended-instrument col 3 fails the over-identification test, likely reflecting one of the additional extended-controls W_j violating the heteroskedasticity-mean-independence exclusion restriction. The Wu-Hausman test of OLS-consistency under exogeneity does not reject the null in col 2 ($p = 0.24$); we treat this as a failure to reject rather than evidence in favor of OLS-consistency. The 2SLS-Lewbel point estimate is approximately five times the OLS estimate, a magnitude consistent with classical measurement-error attenuation in OLS; the 2SLS-Lewbel and OLS point estimates are not statistically distinguishable from each other at conventional thresholds given the wider 2SLS standard error.

4.3.5 ClarityCEO vs. UncResCEO: Identification Asymmetry

The three designs target different speech components by construction. The Phase E and DWZ first-difference designs identify ClarityCEO via cross-CEO-pair variation around turnover events; neither identifies UncResCEO, because the residual of a CEO-fixed-effect regression has within-CEO mean exactly zero by OLS first-order conditions, so any first-difference that averages within-CEO has $\Delta\text{UncResCEO} \equiv 0$ in expectation. The Lewbel design identifies UncResCEO via call-level heteroskedasticity-based instruments; it does not separately identify ClarityCEO because ClarityCEO is a CEO-level trait (one observation per CEO in the panel) whose call-level variation is, by construction, the variation absorbed into UncResCEO. The two regressors therefore have structurally distinct identification problems and require structurally distinct identification tools. The asymmetry across designs reflects the underlying statistical structure of the two speech components, not a coverage gap in the endogeneity package.

5 Discussion and Conclusion

5.1 Summary of Findings

This paper extends the three-component CEO speech-uncertainty decomposition of Dzieliński et al. (2021)—persistent CEO style `ClarityCEO`, call-level Q&A residual `UncResCEO`, and presentation-segment uncertainty share `UncPreCEO`—from the market-reaction, analyst-behavior, firm-performance, and CEO-compensation outcome set on which Dzieliński et al. validate it to a class of outcomes their paper does not test: firm-level financing-policy decisions and outsider regulatory reaction. On a panel of 112,968 earnings-call transcripts covering 2,429 U.S. firms over fiscal years 2002–2018, four empirical patterns emerge.

First, the Q&A residual `UncResCEO` loads positively on cash holdings in 12 of 12 fixed-effects specifications at $p < 0.10$ one-tailed, with point estimates between +0.0016 and +0.0046 across the Full and QtrExp measurement methods. Persistent CEO style `ClarityCEO` loads negatively in 9 of 12 specifications (5 of 12 at $p < 0.05$ in Full; 9 of 12 at $p < 0.05$ and 6 of 12 at $p < 0.01$ in QtrExp). `UncPreCEO` is null on the cash margin once the Q&A components are jointly included. Second, the cash response amplifies among credit-constrained firms: the `UncResCEO` × Unrated interaction is positive at $p < 0.05$ in 2 of 8 cells on the lead-DV specifications, and the `UncResCEO` × HighCFvol interaction reproduces the asymmetry through the cash-flow-volatility moderator at one tier weaker. Third, all four uncertainty drivers (firm political risk, U.S. economic-policy uncertainty, global economic-policy uncertainty, and product-market competition) load positively on the speech-uncertainty constructs, establishing construct validity for the decomposition before the firm-side outcome tests. Fourth, the post-call quoted bid-ask spread and SEC comment-letter receipt both load on `UncPreCEO` (4 of 12 spread cells and 4 of 6 SEC cells significant at $p < 0.10$); the Q&A components are null on both outsider-reaction outcomes.

5.2 Channel-Asymmetry Synthesis

The four empirical patterns share a single structural interpretation. The firm's own financing-policy response (Section 3) loads on the Q&A components of CEO speech; the outsider-reaction outcomes (Section 4.2) load on the presentation segment. The asymmetry sits cleanly on Dzieliński et al.'s own theoretical structure for the two segments of the call: the presentation segment is the one Dzieliński et al. characterize as "scripted and vetted beforehand… likely to reflect the firm's culture", and the Q&A segment is the one they characterize as "inevitably somewhat improvised and hence more likely to reveal each CEO's style." Two segments of the same call carry signals to actors with different information sets and different decision horizons. Insiders making the firm's financing decisions track Q&A signals—both the persistent CEO trait and the call-level state residual; outsiders running market-microstructure and regulatory-review processes track the presentation share. We document the asymmetry within Dzieliński et al.'s theoretical scaffolding rather than impose it.

5.3 Mechanism Disclosure

The Section 3 firm-side cash response and the Section 4.2 outsider-reaction responses operate through theoretically distinct mechanisms; we do not claim a causal bridge between them. The firm-side response is anchored in the precautionary motive of Opler et al. (1999) and Bates et al. (2009): uncertainty raises optimal cash, with the call-level Q&A residual a real-time within-call uncertainty signal. The outsider-reaction response is anchored in information-asymmetry and regulatory-review processes (Bushee et al. (2018) for market microstructure; Lerman et al. (2026) for SEC scrutiny), where the scripted prepared-remarks share is the natural object outsiders parse against the firm's mandatory filings. The two mechanisms can hold simultaneously without either causing the other: the same call carries Q&A signals to insiders and presentation signals to outsiders, and both populations respond in the directions their respective theoretical models predict.

Section 4.3 addresses three identification threats to the firm-side cash response with three orthogonal designs: a CEO sudden-death difference-in-differences against the reverse-causality threat, a first-difference replication of Dzieliński et al.'s Section 6 against time-invariant heterogeneity, and a Lewbel (2012) heteroskedasticity-based instrumental-variable estimator against time-varying omitted confounders. The three designs target different identification threats and different speech components by construction: the Phase E and DWZ first-difference designs identify the persistent-style component; the Lewbel design identifies the call-level residual. The asymmetry reflects the underlying statistical structure of the two speech components, not a coverage gap.

5.4 Limitations

Three limitations qualify the empirical evidence. First, the Phase E sudden-death sample size of eight viable events is below conventional power thresholds; the four DiD specifications report negative point estimates in the predicted direction but none statistically significant at $p < 0.10$ one-tailed, and the originally proposed heterogeneity test on pre-event speech-uncertainty composition is not feasible. Second, the Lewbel (2012) heteroskedasticity-based estimator delivers a 2SLS point estimate approximately five times the OLS estimate but with wider standard errors; the Wu-Hausman test does not reject OLS-consistency, and we treat this as a failure to reject rather than evidence in favor of OLS-consistency. Third, the cash dependent variable is the linear cash-to-assets ratio of Bates et al.; the log-of-cash-to-net-assets specification of Opler et al. is flagged as a planned robustness extension but not estimated in the present draft.

5.5 Future Research

Four directions extend this paper's results. First, the Phase E sudden-death DiD heterogeneity test by pre-event `UncResCEO` composition is naturally extended to a larger event sample. Second, the Opler et al. log-of-cash-to-net-assets robustness is a one-runner extension reported in the body's robustness section. Third, the firm-side response to CEO speech uncertainty is documented at the cash margin; extension to other financing-policy margins (leverage, payout, capital structure) under the same three-component decomposition is a natural continuation. Fourth, the channel asymmetry between insider and outsider reactions admits sharpening through experimental or quasi-experimental settings that vary the segment of the call exogenously; we leave this as a methodological direction for the disclosure-quality literature.

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A Variable Definitions

This appendix lists every variable used in the empirical analyses, with construction formula, data source, and originating reference. Variable names follow paper-standard conventions where available (Dzieliński et al. 2021 for speech measures, Bates et al. 2009 for cash-holdings controls); deviations from source-paper frequency or sample are documented in the formula column.

A.1 Sample Manifest

Name	Description / Formula	Source	Reference
`file_name, ceo_id, ceo_name, gvkey, ff12_code, ff12_name, start_date`	Master sample manifest with CEO, firm, and industry identifiers	outputs/1.4_AssembleManifest	

A.2 Text/Linguistic Variables

Name	Description / Formula	Source	Reference
`ClarityCEO`	CEO clarity persistent-style component (DWZ Eqn 5). *Formula:* $\text{ClarityCEO}_i = -\widehat{\text{FE}}_i$ where $\widehat{\text{FE}}_i$ is the CEO-i fixed-effect estimate from the panel regression $\text{UncAnsCEO}_{ic} = \alpha + \sum_i \text{FE}_i \cdot \mathbb{1}[\text{CEO} = i] + \mathbf{X}_{ic}\gamma + \varepsilon_{ic}$ over the call-level Q&A uncertainty panel; controls $\mathbf{X}$ are the DWZ extended-controls vector. The sign flip on $\widehat{\text{FE}}_i$ makes higher ClarityCEO mean "CEO speaks more clearly" (lower mean uncertainty). Persistent CEO trait.	outputs/econometric/ceo_clarity_extended/	Dzielínski et al. (2021) (Section 4.4 Eqn 5, p.16)
`ClarityCEO_QtrExp`	ClarityCEO under within-tenure expanding window. *Formula:* As ClarityCEO, but the CEO fixed-effect regression is re-estimated on an expanding within-CEO window using only calls of CEO i available up to and including focal call c. Strictly look-ahead-free.	outputs/econometric/ceo_clarity_expanding/	thesis QtrExp variant of Dzielínski et al. (2021) Eqn 5
`NegCall`	Entire-call negative sentiment percentage. *Formula:* Ratio of negative words to total words in the entire call, based on LM 2011 negative-words list. Per DWZ 2021 Appendix Table A.1: 'percentage of negative words in all words spoken by the CEO, CFO and analysts attending the call.'	outputs/2_Textual_Analysis/2.2_Variables	Dzielínski et al. (2021) (Section 4.4, p.18; Appendix Table A.1, p.54)
`UncAnsCEO`	CEO-only Q&A uncertainty percentage (input to ClarityCEO/UncResCEO residualization). *Formula:* CEO Q&A uncertainty (%) = UnctWordsAnsCEO / WordsAnsCEO (DWZ 2021 Eqn 2). Uncertainty wordlist = LM 2011 Master Dictionary 'uncertainty' list (297 words, Aug 2014 version). CEO identified via Execucomp ceoann field. Used in body only as the dependent variable inside the CEO fixed-effect residualization that produces ClarityCEO and UncResCEO.	outputs/2_Textual_Analysis/2.2_Variables	Dzielínski et al. (2021) (Section 4.2, Eqn 2, p.13)
`UncPreCEO`	CEO-only presentation-segment uncertainty percentage (DWZ Eqn 1). *Formula:* CEO Presentation uncertainty (%) = UnctWordsPreCEO / WordsPreCEO. Uncertainty wordlist = LM 2011 (297 words).	outputs/2_Textual_Analysis/2.2_Variables	Dzielínski et al. (2021) (Section 4.2, Eqn 1, p.13)
`UncQue`	Analyst Q&A uncertainty percentage (DWZ Eqn 3). *Formula:* Analyst Q&A uncertainty (%) = UnctWordsQue / WordsQue. Uncertainty wordlist = LM 2011 (297 words). Used as ΔUncQue control in DWZ first-difference replication.	outputs/2_Textual_Analysis/2.2_Variables	Dzielínski et al. (2021) (Section 4.2, Eqn 3, p.13)

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Name	Description / Formula	Source	Reference
`UncResCEO`	CEO Q&A residual call-state uncertainty component (DWZ Eqn 4). *Formula:* $\widehat{\varepsilon}_{ic}$ = the residual from the panel regression UncAnsCEO$_{ic}$ = $\alpha + \sum_i \mathrm{FE}_i \cdot \mathbb{1}[\mathrm{CEO} = i] + \mathbf{X}_{ic}\boldsymbol{\gamma} + \varepsilon_{ic}$ that produces ClarityCEO. By OLS first-order conditions, the within-CEO mean of UncResCEO is exactly zero by construction. Captures call-level deviation in CEO Q&A uncertainty after stripping out the persistent CEO mean.	outputs/econometric/ceo_clarity_extended/	Dzieliński et al. (2021) (Section 4.4 Eqn 4, p.16)
`UncResCEO_QtrExp`	UncResCEO under within-tenure expanding window. *Formula:* As UncResCEO, but the residualization regression is re-estimated on an expanding within-CEO window. Strictly look-ahead-free.	outputs/econometric/ceo_clarity_expanding/	thesis QtrExp variant of Dzieliński et al. (2021) Eqn 4

A.3 Financial / Market Variables

Name	Description / Formula	Source	Reference
`BGTAvg_Amihud`	BGT 25-day symmetric average Amihud illiquidity (H7e). *Formula:* Mean daily Amihud illiquidity over [-25, +25] trading days around call (51-day symmetric window, day 0 INCLUDED). F1D extension to BGT 2018.	.../variables/bgt_long_window_amihud.py	bgt2018 (window); F1D extension (symmetric average shape)
`BGTAvg_Spread`	BGT 25-day symmetric average bid-ask spread (H14e). *Formula:* Mean daily relative bid-ask spread over [-25, +25] trading days around call (51-day symmetric window, day 0 INCLUDED). Spread = (ASK - BID) / ((ASK + BID) / 2). F1D extension to BGT 2018.	.../variables/bgt_long_window_spread.py	bgt2018 (window); F1D extension (symmetric average shape)
`BGTDelta_Amihud`	BGT 25-day post-pre Amihud illiquidity change (H7d). *Formula:* Mean Amihud over post window [+1, +25] minus mean over pre window [-25, -1] trading days (day 0 EXCLUDED). F1D extension to BGT 2018.	.../variables/bgt_long_window_amihud.py	bgt2018 (window); F1D extension (delta shape)
`BGTDelta_Spread`	BGT 25-day post-pre bid-ask spread change (H14d). *Formula:* Mean spread over [+1, +25] minus mean over [-25, -1] trading days (day 0 EXCLUDED). F1D extension to BGT 2018.	.../variables/bgt_long_window_spread.py	bgt2018 (window); F1D extension (delta shape)
`BGTLevel_Amihud`	Compute BGT (2018) long-window Amihud illiquidity around earnings calls. *Formula:* CRSP via get_raw_daily_data (see module: src/f1d/shared/variables/bgt_long_window_amihud.py)	CRSPviaget_raw_daily_data	Bushee, Gow & Taylor (2018, JAR) 56(1):85-121
`BGTLevel_Spread`	Compute BGT (2018) long-window closing bid-ask spread around earnings calls. *Formula:* CRSP via get_raw_daily_data (closing BID/ASK) (see module: src/f1d/shared/variables/bgt_long_window_spread.py)	CRSPviaget_raw_daily_data	Bushee, Gow & Taylor (2018, JAR) [window]; Lee (2016, TAR) [formula]
`Capex`	Build Capex = capxy_annual / atq_lag from raw Compustat quarterly data. *Formula:* Compustat/capxy_Q4/atq (see module: src/f1d/shared/variables/capex_intensity.py)	Compustat/capxy_Q4/atq	Bates, Kahle & Stulz (2009, JF)
`CashFlowAt`	Build CashFlow = oancfy / atq from raw Compustat quarterly data. *Formula:* Compustat/oancfy/atq (see module: src/f1d/shared/variables/cash_flow.py)	Compustat/oancfy/atq	Bates, Kahle & Stulz (2009, JF)

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Name	Description / Formula	Source	Reference
`CashRatio`	Build CashRatio = cheq / atq from raw Compustat quarterly data. *Formula:* Compustat/cheq/atq (see module: src/f1d/shared/variables/cash_holdings.py)	Compustat/cheq/atq	Bates et al. (2009) (Section II, p.1991; BKS use annual che/at — we use quarterly cheq/atq for panel alignment)
`ChangDebtChoice`	ChangDebtChoice (description not in module docstring) *Formula:* see module: src/f1d/shared/variables/_compustat_engine.py	.../variables/_compustat_engine.py	cdh2006
`ChangExternalFunding`	ChangExternalFunding (description not in module docstring) *Formula:* see module: src/f1d/shared/variables/_compustat_engine.py	.../variables/_compustat_engine.py	cdh2006
`DailyVola`	Build Volatility from raw CRSP daily stock files via CRSPEngine. *Formula:* CRSP/RET (see module: src/f1d/shared/variables/volatility.py)	CRSP/RET	dwz2021
`DebtToCapital`	Build DebtToCapital = total debt / (equity + total debt). *Formula:* Compustat/dlcq,dlttq,seqq (see module: src/f1d/shared/variables/debt_to_capital.py)	Compustat/dlcq,dlttq,seqq	rz1995
`DivDummy`	Build DivDummy = (dvy ¿ 0).astype(float) from raw Compustat quarterly data. *Formula:* Compustat/dvy¿0 (see module: src/f1d/shared/variables/dividend_payer.py)	Compustat/dvy¿0	Bates et al. (2009) (Section IV, p.2000; dividend payout dummy based on common dividends)
`DivPayerQ`	Build DivPayerQ = (dvpsxq ¿ 0).astype(float) from raw Compustat quarterly data. *Formula:* Compustat/dvpsxq¿0 (quarterly ex-date) (see module: src/f1d/shared/variables/dividend_payer_quarterly.py)	Compustat/dvpsxq¿0	Hoberg & Prabhala (2009, RFS) — Compustat item 26 (dvpsx) analog
`EarnVol`	Builder for Earnings Volatility (earnings_volatility) variable. *Formula:* Compustat (see module: src/f1d/shared/variables/earnings_volatility.py)	Compustat	Duong, Do, and Do (2025) (Appendix B, p.30; 3-year rolling std of qoq asset-scaled earnings)
`EquityDelayCon`	Hoberg-Maksimovic (2015) equity-market financing constraint index (H22). *Formula:* Hoberg-Maksimovic (2015) firm-year equity-market financing constraint measure. Higher values = more constrained firm. Panel column is lowercase 'equitydelaycon'; CamelCase name is the spec convention. See docs/provenance/h22.md for merge details.	inputs/Hoberg_Maksimovic/	hm2015
`FirmMat`	Builder for Firm Maturity (firm_maturity) variable. *Formula:* Compustat (see module: src/f1d/shared/variables/firm_maturity.py)	Compustat	Biddle, Hilary, and Verdi (2009) (Appendix A, p.130; years since first CRSP appearance); alt Leary and Roberts (2010) (App.~A p.354)
`Leverage`	Build Leverage = (dlcq + dlttq) / atq (interest-bearing debt only). *Formula:* Compustat/dlcq,dlttq,atq (see module: src/f1d/shared/variables/book_lev.py)	Compustat/dlcq,dlttq,atq	bks2009, rz1995
`lnAssets`	Build Size = ln(atq) from raw Compustat quarterly data. *Formula:* Compustat/atq (see module: src/f1d/shared/variables/size.py)	Compustat/atq	dwz2021
`Loss`	Builder for Loss (H5 Control Variable). *Formula:* Compustat ibq ¡ 0 (see module: src/f1d/shared/variables/loss_dummy.py)	Compustatibq¡0	Biddle et al. (2009) (Appendix A, p.131; indicator for negative net income before extraordinary items)

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Name	Description / Formula	Source	Reference
PayoutRatio_q	Build PayoutRatio_q = (dvpspq × cshoq) / ibq from Compustat. *Formula:* Compustat/(dvpspq*cshoq)/ibq (quarterly) (see module: src/f1d/shared/variables/payout_ratio_quarterly.py)	Compustat/	dds2004
PRisk	Match quarterly PRisk onto each earnings call by (gvkey, calendar_quarter). *Formula:* Hassan et al. (2019) quarterly PRisk (see module: src/f1d/shared/variables/prisk_q.py)	Hassanetal.	hhlt2019
RDSales	Build RDSales = xrdq / atq from raw Compustat quarterly data. *Formula:* Compustat/xrdq,atq (see module: src/f1d/shared/variables/rd_intensity.py)	Compustat/xrdq,atq	bks2009, jjl2021
RepurchaseIntensity	Build RepurchaseIntensity = quarterly_prstkcy / lagged_atq from Compustat. *Formula:* Compustat/quarterly_prstkcy/lagged_atq (see module: src/f1d/shared/variables/repurchase_intensity.py)	Compustat/quarterly_prstkcy/lagged_atq	Thesis Advisor Suggestion
ROA	Build ROA = iby_annual / avg_assets from Compustat quarterly data. *Formula:* Compustat/iby,atq (see module: src/f1d/shared/variables/roa.py)	Compustat/iby,atq	Wang (2020), dwz2021
SalesGrowth	Build SalesGrowth from raw Compustat quarterly data (annual Q4 panel). *Formula:* Compustat/saley_Q4_YoY (see module: src/f1d/shared/variables/sales_growth.py)	Compustat/saley_Q4_YoY	Biddle et al. (2009) (Section 3.2, p.117; pct change in sales)
sCFO	Build sCFO = rolling 5-year std (min 3 yrs) of (oancfy/atq_{t-1}) per gvkey. *Formula:* Compustat/oancfy/atq_{t-1} rolling std (see module: src/f1d/shared/variables/ocf_volatility.py)	Compustat/oancfy/atq_{t-1}rollingstd	Biddle et al. (2009) (Appendix A, p.130; 5-year rolling std of CFO/avg assets)
StockPrice	Compute stock price at earnings call date (VECTORIZED). *Formula:* CRSP via get_raw_daily_data (see module: src/f1d/shared/variables/stock_price.py)	CRSPviaget_raw_daily_data	Amiram, Owens, and Rozenbaum (2016) (Appendix A, p.137; daily CRSP stock price; justification cites Stoll 1978)
SurpDec	Signed-quintile earnings surprise rank (-5 to +5) within calendar quarter. *Formula:* Signed quintile rank of percentage earnings surprise per DWZ 2021 Appendix Table A.1: raw surprise = (actual - consensus forecast earnings) / share price 5 trading days before announcement, x 100. Firms grouped into 5 bins of positive surprise (+5 largest through +1 smallest positive), 0 for zero surprises, and 5 bins of negative surprise (-1 smallest negative through -5 largest negative). DWZ footnote 32 notes this decile convention follows Hirshleifer, Lim & Teoh (2009) and DellaVigna & Pollet (2009).	.../variables/earnings_surprise.py	Dzieliński et al. (2021) (Appendix Table A.1, p.55; convention cites DellaVigna-Pollet 2009 + Hirshleifer-Lim-Teoh 2009)
TobinsQ	Build TobinsQ = (AT + cshoq*prccq - CEQ) / AT from raw Compustat quarterly data. *Formula:* Compustat/(atq+cshoq*prccq-ceqq)/atq (see module: src/f1d/shared/variables/tobins_q.py)	Compustat/	Bates, Kahle & Stulz (2009, JF)
Turnover	Compute share turnover at earnings call date (VECTORIZED). *Formula:* CRSP via get_raw_daily_data (see module: src/f1d/shared/variables/turnover.py)	CRSPviaget_raw_daily_data	Amihud (2002) (Section 2.1, p.33; alt rolling-window def in Chang, Dasgupta, and Hilary (2006) Appendix B p.3045)

A.4 Econometric / Indicator Variables

Name	Description / Formula	Source	Reference
BelowIG	Below-IG credit rating subsample indicator (H1.2). *Formula:* Binary = 1 if firm's S&P long-term issuer credit rating is below investment grade (BB+ and below) at call date; 0 otherwise. Rated firms only (Unrated firms get separate indicator).	runtime	thesis (H1.2 rating subsample)
HighCFvol	High cash-flow-volatility moderator (H1.3). *Formula:* Binary = 1 if firm-quarter Han-Qiu (2007) cash-flow-volatility (16-quarter rolling std of OCF / |mean| of OCF) is above the within-industry-year median; 0 otherwise.	runtime	Han and Qiu (2007); thesis (H1.3 split convention)
HighTSIMM	High product-market-similarity indicator (H1.1). *Formula:* Binary = 1 if firm-year TotalSimilarity (HP 2016 product market similarity) is above the within-year median; 0 otherwise.	runtime	hp2016 (TSIMM construction); thesis (H1.1 split convention)
Lagged_DV	Generic lagged-DV control (1Q lag of suite-specific DV). *Formula:* 1-quarter lag of the dependent variable, constructed at runtime per spec (DV varies per suite). Always included as base control to absorb persistence.	runtime	thesis convention (per feedback_lagged_dv.md)
Unrated	No-rating subsample indicator (H1.2). *Formula:* Binary = 1 if firm has no S&P rating at call date; 0 otherwise.	runtime	thesis (H1.2 rating subsample)
z_log_TotalSimilarity	z-scored log of HP 2016 TotalSimilarity (continuous moderator, H1.1). *Formula:* z-score of log(1 + TotalSimilarity) computed within year. TotalSimilarity = HP 2016 firm-year product market similarity score (sum of pairwise cosine similarities × 100).	runtime	hp2016

A.5 Runtime Derivatives (Lead/Lag/Centered/Interaction)

Name	Description / Formula	Source	Reference
AbsSurpDec	Absolute earnings surprise quintile magnitude. *Formula:* abs(SurpDec) — unsigned magnitude of signed-quintile earnings surprise rank, values 0-5.	runtime	thesis (magnitude control independent of sign)
BGTAvg_Amihud_lead1	1Q lead of BGTAvg_Amihud. *Formula:* 1-quarter lead of BGTAvg_Amihud	runtime	bgt2018 (window); F1D extension (symmetric average shape)
BGTAvg_Spread_lead1	1Q lead of BGTAvg_Spread. *Formula:* 1-quarter lead of BGTAvg_Spread	runtime	bgt2018 (window); F1D extension (symmetric average shape)
BGTDelta_Amihud_lead1	1Q lead of BGTDelta_Amihud. *Formula:* 1-quarter lead of BGTDelta_Amihud	runtime	bgt2018 (window); F1D extension (delta shape)
BGTDelta_Spread_lead1	1Q lead of BGTDelta_Spread. *Formula:* 1-quarter lead of BGTDelta_Spread	runtime	bgt2018 (window); F1D extension (delta shape)
BGTLevel_Amihud_lead1	1Q lead of BGTLevel_Amihud. *Formula:* 1-quarter lead of BGTLevel_Amihud	runtime	bgt2018

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Name	Description / Formula	Source	Reference
BGTLevel_Spread_lead1	1Q lead of BGTLevel_Spread. *Formula:* 1-quarter lead of BGTLevel_Spread	runtime	bgt2018, lee2016
Capex_lead	1Q lead of Capex. *Formula:* 1-quarter lead of Capex	runtime	Bates, Kahle & Stulz (2009, JF)
Capex_lead2	2Q lead of Capex. *Formula:* 2-quarter lead of Capex	runtime	Bates, Kahle & Stulz (2009, JF)
Capex_lead3	3Q lead of Capex. *Formula:* 3-quarter lead of Capex	runtime	Bates, Kahle & Stulz (2009, JF)
Capex_lead4	4Q lead of Capex. *Formula:* 4-quarter lead of Capex	runtime	Bates, Kahle & Stulz (2009, JF)
CashRatio_lead	1Q lead of CashRatio. *Formula:* 1-quarter lead of CashRatio	runtime	see base var
ChangDebtChoice_lead	1Q lead of ChangDebtChoice. *Formula:* 1-quarter lead of ChangDebtChoice	runtime	cdh2006
ChangExternalFunding_lead	1Q lead of ChangExternalFunding. *Formula:* 1-quarter lead of ChangExternalFunding	runtime	cdh2006
DebtToCapital_lead	1Q lead of DebtToCapital. *Formula:* 1-quarter lead of DebtToCapital	runtime	rz1995
DeltaILLIQ_lead1	1Q lead of DeltaILLIQ. *Formula:* 1-quarter lead of DeltaILLIQ	runtime	amihud02
DISP_lag	1Q lag of DISP. *Formula:* 1-quarter lag of DISP	runtime	Wang (2020)
DISP_lead	1Q lead of DISP. *Formula:* 1-quarter lead of DISP	runtime	Wang (2020)
DivPayerQ_lead1	1Q lead of DivPayerQ. *Formula:* 1-quarter lead of DivPayerQ	runtime	Hoberg & Prabhala (2009, RFS) – primary; Chetty & Saez (2005, QJE) – secondary (firm-quarter frequency precedent)
DSPREAD_lead1	1Q lead of DSPREAD. *Formula:* 1-quarter lead of DSPREAD	runtime	lee2016
EquityDelayCon_lead	1Q lead of EquityDelayCon. *Formula:* 1-quarter lead of EquityDelayCon	runtime	hm2015
Leverage_lead	1Q lead of Leverage. *Formula:* 1-quarter lead of Leverage	runtime	bks2009, rz1995

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Name	Description / Formula	Source	Reference
PayoutRatio_q_lead_qtr	1Q lead of PayoutRatio_q (qtr-aligned). *Formula:* 1-quarter lead of PayoutRatio_q (calendar-quarter aligned)	runtime	dds2004
PostCallAmihud	Panel-time post-call Amihud illiquidity level (H7b). *Formula:* Panel-time mean daily Amihud illiquidity over the post-call period (call quarter + N following quarters per H7b runner spec). Computed on panel structure with two-way clustering.	runtime	thesis (H7b panel-time post-call construction)
PostCallAmihud_lead1	1Q lead of PostCallAmihud. *Formula:* 1-quarter lead of PostCallAmihud	runtime	thesis (H7b panel-time post-call construction)
PostCallSpread	Panel-time post-call bid-ask spread level (H14b). *Formula:* Panel-time mean daily relative bid-ask spread over the post-call period (call quarter + N following quarters per H14b runner spec). Computed on panel structure with two-way clustering.	runtime	thesis (H14b panel-time post-call construction)
PostCallSpread_lead1	1Q lead of PostCallSpread. *Formula:* 1-quarter lead of PostCallSpread	runtime	thesis (H14b panel-time post-call construction)
PRisk_lag	1Q lag of PRisk. *Formula:* 1-quarter lag of PRisk	runtime	hhlt2019
PRisk_lag2	2Q lag of PRisk. *Formula:* 2-quarter lag of PRisk	runtime	hhlt2019
RDSales_lead	1Q lead of RDSales. *Formula:* 1-quarter lead of RDSales	runtime	bks2009, jjl2021
RepurchaseIntensity_lead_qtr	1Q lead of RepurchaseIntensity (qtr-aligned). *Formula:* 1-quarter lead of RepurchaseIntensity (calendar-quarter aligned)	runtime	Thesis Advisor Suggestion
ClarityCEO_c	Mean-centered ClarityCEO. *Formula:* ClarityCEO minus its sample mean (mean-centered for interaction interpretability)	runtime	thesis (interaction interpretability)
ClarityCEO_QtrExp_c	Mean-centered ClarityCEO_QtrExp. *Formula:* ClarityCEO_QtrExp minus its sample mean	runtime	thesis (interaction interpretability)
Post	DiD post-event-period indicator (Phase E sudden-death). *Formula:* Binary = 1 if call-quarter is on or after the death-event quarter for the treated firm (and on or after the matched event-quarter for the matched-control firm); 0 otherwise.	runtime	Ghafoor et al. (2023)
Treated	DiD treated-firm indicator (Phase E sudden-death). *Formula:* Binary = 1 if firm experienced a CEO sudden-death event in 2002–2018; 0 if PSM-matched control firm.	runtime	Bennedsen et al. (2020); Ghafoor et al. (2023)
Treated_x_Post	DiD ATT term (Phase E sudden-death). *Formula:* Treated $\times$ Post. The coefficient on this product is the average treatment effect on the treated.	runtime	Bertrand et al. (2004); Ghafoor et al. (2023)

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Name	Description / Formula	Source	Reference
UncPreCEO_c	Mean-centered UncPreCEO. *Formula:* UncPreCEO minus its sample mean (mean-centered for interaction interpretability)	runtime	thesis (interaction interpretability)
UncPreCEO_c_x_HighCFvol	Interaction: UncPreCEO_c × HighCFvol (H1.3). *Formula:* Product: UncPreCEO_c * HighCFvol	runtime	thesis (CFvol moderator interaction)
UncPreCEO_c_x_Unrated	Interaction: UncPreCEO_c × Unrated (HFC). *Formula:* Product: UncPreCEO_c * Unrated	runtime	thesis (HFC interaction)
UncResCEO_c	Mean-centered UncResCEO. *Formula:* UncResCEO minus its sample mean (mean-centered for interaction interpretability)	runtime	thesis (interaction interpretability)
UncResCEO_c_x_HighCFvol	Interaction: UncResCEO_c × HighCFvol (H1.3). *Formula:* Product: UncResCEO_c * HighCFvol	runtime	thesis (CFvol moderator interaction)
UncResCEO_c_x_Unrated	Interaction: UncResCEO_c × Unrated (HFC). *Formula:* Product: UncResCEO_c * Unrated	runtime	thesis (HFC interaction)
UncResCEO_QtrExp_c	Mean-centered UncResCEO_QtrExp. *Formula:* UncResCEO_QtrExp minus its sample mean	runtime	thesis (interaction interpretability)
UncResCEO_QtrExp_c_x_Unrated	Interaction: UncResCEO_QtrExp_c × Unrated (HFC QtrExp). *Formula:* Product: UncResCEO_QtrExp_c * Unrated	runtime	thesis (HFC QtrExp interaction)

A.6 Stage 9

Name	Description / Formula	Source	Reference
CCCL	SEC comment letter received (binary, forward-looking) *Formula:* 1 if SEC comment letter received in next quarter, 0 otherwise	inputs/EDGAR_CommentLetter s	cdm2013
DeltaILLIQ	Change in Amihud illiquidity around earnings call *Formula:* PostCallILLIQ - PreCallILLIQ over [+1,+3] vs [-3,-1] days	inputs/CRSP_DSF	amihud02
DISP	Analyst forecast dispersion (Wang 2020) *Formula:* SD(latest analyst EPS forecasts in T-31..T-1 window) / prccq_prior	inputs/tr_ibes	Wang (2020)
DSPREAD	Change in closing bid-ask spread around earnings call *Formula:* mean(RelSpread[+1,+3]) - mean(RelSpread[-3,-1]) where RelSpread = 2*(ASK-BID)/(ASK+BID)	inputs/CRSP_DSF	lee2016
GEPU_log	Davis (2016, NBER WP 22740) Global Economic Policy Uncertainty (GEPU) index (log-transformed, current-$ GDP weights) *Formula:* log(GEPU_current) — GDP-weighted average of 16 country EPU indices, matched by calendar month of call start_date	inputs/EconomicPolicyUncertai ntyIndex/Global	Davis (2016)

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Name	Description / Formula	Source	Reference
`GPR_log`	Caldara & Iacoviello (2022, AER) headline Geopolitical Risk index (log-transformed, 10-newspaper recent version) *Formula:* log(GPR) — recent headline Geopolitical Risk index, matched by calendar month of call start_date	inputs/matteoiacoviello/GPR_Gl obal	ci2022
`SEC_Letters_fwd`	SEC comment letter count (forward-looking) *Formula:* Count of SEC-originated letters (UPLOAD type) in next calendar quarter	inputs/EDGAR_CommentLetter s	cdm2013 (thesis extension to count)
`TotalSimilarity`	Hoberg-Phillips total product similarity *Formula:* Sum of pairwise cosine similarities from 10-K product descriptions	inputs/TNIC	hp2016
`US_EPU_log`	Baker, Bloom & Davis (2016, QJE) US News-Based Economic Policy Uncertainty index (log-transformed) *Formula:* log(News_Based_Policy_Uncert_Index) matched by calendar month of call start_date	inputs/EconomicPolicyUncertai ntyIndex/US	bbd2016

B Robustness Specifications

This appendix collects auxiliary specifications and design-choice disclosures supporting the main §3 hypotheses HC and HFC.

B.1 Binary Constraint Moderator: Rated-versus-Unrated

The HFC specification of §3.3 adopts the binary rated-versus-unrated partition of Faulkender and Petersen (2006) directly: the Unrated indicator equals one for firms without a Compustat credit rating, and the rated reference category pools investment-grade and below-investment-grade firms. The binary scheme follows Faulkender and Petersen verbatim and is consistent with the macro-shock asymmetry framing of Almeida et al. (2004) §II.D. We do not estimate a three-tier IG/BelowIG/Unrated specification: the main hypothesis localizes the precautionary cash response to the credit-constrained Unrated cell, and the rated half of the population is heterogeneous in ways that the binary scheme treats as nuisance variation absorbed into the rated-pool reference.

B.2 Alternative Cash Dependent-Variable Specification (OPSW Log-Form)

The body adopts the linear cash-to-assets dependent variable (`cheq/atq`) of Bates et al. (2009) as the standard precautionary-cash form. Opler et al. (1999) also use a log-form dependent variable, defined as $\log(\text{cheq}/\text{atq})$ (or, in some specifications, $\log(\text{cheq}/(\text{atq} - \text{cheq}))$ to avoid log-of-zero issues at the lower tail). The log form down-weights the upper tail of the cash distribution relative to the linear form and tests whether the precautionary association is driven by high-cash firms.

The present draft does not estimate the OPSW log-DV robustness; it is flagged in §5 as a planned extension. The expectation, given the magnitude of the linear-form coefficients in §3.2 (`UncResCEO` significant 12 of 12 at $p < 0.10$), is that the qualitative result survives under the log specification, with attenuated point-estimate magnitudes due to the tail-compression. Future work will report the log-DV coefficients directly to confirm this expectation.

B.3 Fixed-Effects Ladder Robustness

The four-step fixed-effects ladder of §2.5 (Industry+Year, Firm+Year, Industry+Year-Quarter, Firm+Year-Quarter) is the within-regressor robustness frame applied throughout the body. Every column of every body table reports a different point on this ladder, so the FE-ladder robustness check is integrated rather than a separate appendix specification. The convergence pattern across the ladder—attenuation from industry to firm fixed effects, additional attenuation from year to year-quarter fixed effects—is the standard within-firm panel-OLS attenuation pattern (within-firm variation is a strict subset of cross-firm variation, so coefficients on a regressor with substantial across-firm dispersion shrink as fixed effects absorb that dispersion). The signs and significance of `UncResCEO` are robust to all four ladder steps in HC, HFC, and the cash-flow-volatility moderator; this robustness is documented in the body and not separately tabulated here.

B.4 Full and QtrExp Measurement Methods

The body reports each of the HC, HFC, and CFvol suites under two parallel measurement methods. The Full method estimates the underlying CEO fixed-effect regression on the entire firm-quarter panel and recovers the negative of the CEO fixed effect as `ClarityCEO` and the regression residual as `UncResCEO` (the original Dzielínski et al. (2021) construction). The QtrExp method estimates an expanding-window CEO fixed effect on within-CEO data only, using only the calls of a given CEO available up to and including the focal call. The QtrExp method is strictly look-ahead-free: the recovered components for a focal call use no information from calls that follow it. The two methods agree in direction on all three speech components in all body suites and yield slightly stronger statistical evidence for the persistent-style `ClarityCEO` component in the QtrExp method (which uses fewer observations and a more demanding within-CEO identification window). The agreement between methods supports interpreting the recovered components as CEO-trait and call-state signals rather than as panel-fitted artifacts.

Table 7: Summary Statistics

Variable	Unit	N	Mean	SD	Min	P25	Median	P75	Max
A. Independent Variables									
ClarityCEO	C	61,626	0.2143	0.2129	-0.7717	0.0952	0.2365	0.3629	0.8614
ClarityCEO_QtrExp	C	25,520	0.2011	0.2281	-1.0232	0.0713	0.2256	0.3589	0.8694
GEPU_log	C	88,205	4.6872	0.3672	3.8626	4.3797	4.7238	4.9666	5.5106
PRisk	C	85,995	99.6187	146.5189	0.0000	17.0778	53.3729	120.1973	1192.4308
PRisk_lag2	C	80,627	99.9614	147.2392	0.0000	17.1702	53.4446	120.2666	1192.4308
TotalSimilarity	F	22,403	3.4277	5.6522	1.0000	1.1246	1.5865	3.0903	80.1876
US_EPU_log	C	88,205	4.7123	0.3661	3.8018	4.4805	4.6891	4.9625	5.6478
UncPreCEO	C	71,727	0.6602	0.3917	0.0000	0.3810	0.6001	0.8723	2.2432
UncResCEO	C	44,900	-0.0000	0.3010	-1.5108	-0.1846	-0.0180	0.1650	1.7794
UncResCEO_QtrExp	C	25,520	-0.0057	0.3208	-1.6084	-0.2028	-0.0210	0.1707	1.7402
Unrated	C	79,487	0.5159	0.4997	0.0000	0.0000	1.0000	1.0000	1.0000
B. Dependent Variables									
BGTLevel_Spread	C	87,018	0.0017	0.0026	0.0001	0.0004	0.0009	0.0017	0.0164
BGTLevel_Spread_lead1	C	82,843	0.0016	0.0024	0.0001	0.0004	0.0009	0.0016	0.0164
CCCL	C	88,205	0.0032	0.0563	0.0000	0.0000	0.0000	0.0000	1.0000
CashRatio	C	87,990	0.1708	0.1806	0.0000	0.0368	0.1047	0.2435	0.9938
CashRatio_lead	C	82,236	0.1715	0.1748	0.0000	0.0407	0.1112	0.2447	0.9938
C. Firm Controls									
AbsSurpDec	C	61,544	2.9300	1.4255	0.0000	2.0000	3.0000	4.0000	5.0000
Capex	C	87,627	0.0487	0.0536	0.0000	0.0176	0.0324	0.0592	0.5203
CashFlowAt	C	87,670	0.1019	0.1147	-3.0324	0.0608	0.1032	0.1526	0.4874
DailyVola	C	82,599	36.7556	21.2437	9.4388	22.9709	31.2762	43.8148	211.9160
DivDummy	C	88,134	0.4755	0.4994	0.0000	0.0000	0.0000	1.0000	1.0000
EarnVol	C	87,440	0.0701	0.1907	0.0006	0.0163	0.0315	0.0695	19.5101
FirmMat	C	87,571	-0.0432	2.1826	-317.5714	-0.0078	0.2191	0.4195	0.9252
Leverage	C	87,994	0.2330	0.2240	0.0000	0.0470	0.2065	0.3455	3.9453
NegCall	C	86,858	0.9210	0.3024	0.0000	0.7038	0.8798	1.0943	2.3901
RDSales	C	87,647	0.1047	0.9342	0.0000	0.0000	0.0048	0.0636	39.7688
ROA	C	87,552	0.0386	0.1330	-3.9826	0.0146	0.0525	0.0925	0.5490
SalesGrowth	C	87,506	0.1173	0.3849	-1.0000	-0.0043	0.0745	0.1762	10.3051
StockPrice	C	87,188	36.8864	32.8119	1.6100	14.8975	28.2400	48.1700	192.0384
TobinsQ	C	87,363	2.1363	1.5197	0.3847	1.2681	1.6791	2.4362	35.6144
Turnover	C	87,188	0.0260	0.0299	0.0007	0.0083	0.0160	0.0310	0.1697
UncQue	C	81,031	1.4398	0.4831	0.0000	1.1216	1.4108	1.7268	3.5945
lnAssets	C	87,994	7.4710	1.6607	-0.5692	6.3008	7.3763	8.5436	12.4592
sCFO	C	83,662	0.0619	0.2233	0.0014	0.0236	0.0395	0.0664	32.4035

Notes: Summary statistics for variables used in the hypothesis tests. Sample period: 2002–2018. Anchor sample (H1, Main): 88,205 earnings-call observations across 1,884 firms. Unit column: C = call-level, F = firm-year. N varies by row because each variable is summarized on its anchor panel's Main sample (complete cases per variable); per-suite samples differ due to lags, control availability, and merges.

Table 8: H1 CEO 3-IV Decomp: ClarityCEO + UncResCEO (DWZ Full) + UncPreCEO

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	CashRatio						CashRatio_lead					
CEO Clarity (DWZ)	**-0.0046***	**-0.0071***	**-0.0058****	-0.0054	**-0.0056****	-0.0052	-0.0052	**-0.0170****	**-0.0068***	**-0.0162****	**-0.0069***	**-0.0161****
	(0.0033)	(0.0051)	(0.0033)	(0.0048)	(0.0033)	(0.0047)	(0.0052)	(0.0095)	(0.0051)	(0.0090)	(0.0051)	(0.0090)
CEO Residual Unc. (DWZ)	**0.0021****	**0.0019****	**0.0020****	**0.0016***	**0.0021****	**0.0016***	**0.0028****	**0.0023****	**0.0026****	**0.0020***	**0.0025****	**0.0019***
	(0.0010)	(0.0010)	(0.0010)	(0.0010)	(0.0010)	(0.0010)	(0.0014)	(0.0013)	(0.0014)	(0.0013)	(0.0014)	(0.0013)
CEO Pres Uncertainty	0.0002	0.0009	0.0003	0.0006	0.0004	0.0007	-0.0003	0.0008	-0.0005	0.0004	-0.0007	0.0002
	(0.0011)	(0.0011)	(0.0011)	(0.0011)	(0.0011)	(0.0011)	(0.0021)	(0.0017)	(0.0021)	(0.0017)	(0.0021)	(0.0017)
Leverage	**-0.0177*****	**-0.0355*****	**-0.0130*****	**-0.0218*****	**-0.0132*****	**-0.0216*****	**-0.0327*****	**-0.0279****	**-0.0274*****	-0.0159	**-0.0275*****	-0.0156
	(0.0048)	(0.0084)	(0.0036)	(0.0082)	(0.0037)	(0.0082)	(0.0120)	(0.0121)	(0.0104)	(0.0118)	(0.0105)	(0.0118)
lnAssets	**-0.0015*****	**-0.0177*****	**-0.0017*****	**-0.0175*****	**-0.0014*****	**-0.0170*****	**-0.0024****	**-0.0268*****	**-0.0020****	**-0.0265*****	**-0.0020****	**-0.0265*****
	(0.0006)	(0.0026)	(0.0005)	(0.0026)	(0.0005)	(0.0026)	(0.0010)	(0.0042)	(0.0009)	(0.0042)	(0.0010)	(0.0042)
TobinsQ	**0.0058*****	**0.0048*****	**0.0042*****	**0.0037*****	**0.0040*****	**0.0034*****	**0.0085*****	0.0025	**0.0056*****	0.0009	**0.0056*****	0.0010
	(0.0006)	(0.0009)	(0.0007)	(0.0010)	(0.0007)	(0.0010)	(0.0012)	(0.0015)	(0.0012)	(0.0015)	(0.0012)	(0.0016)
ROA	-0.0144	**0.0214****	**-0.0679*****	-0.0195	**-0.0630*****	-0.0171	**-0.0762*****	**-0.0352***	**-0.1427*****	**-0.0790*****	**-0.1378*****	**-0.0769*****
	(0.0093)	(0.0105)	(0.0113)	(0.0122)	(0.0114)	(0.0122)	(0.0161)	(0.0185)	(0.0173)	(0.0194)	(0.0174)	(0.0196)
Capex	**-0.1117*****	**-0.2995*****	**-0.1596*****	**-0.3074*****	**-0.1564*****	**-0.2993*****	**-0.1613*****	**-0.3281*****	**-0.2433*****	**-0.3415*****	**-0.2411*****	**-0.3430*****
	(0.0128)	(0.0234)	(0.0149)	(0.0236)	(0.0148)	(0.0234)	(0.0184)	(0.0344)	(0.0216)	(0.0345)	(0.0216)	(0.0346)
DivDummy	**-0.0041*****	-0.0028	**-0.0054*****	**-0.0041***	**-0.0049*****	-0.0036	**-0.0051****	-0.0068	**-0.0056****	**-0.0076***	**-0.0054****	**-0.0077***
	(0.0013)	(0.0023)	(0.0013)	(0.0022)	(0.0013)	(0.0022)	(0.0023)	(0.0045)	(0.0023)	(0.0044)	(0.0023)	(0.0044)
sCFO	**0.0066*****	-0.0022	**0.0079*****	-0.0008	**0.0074*****	-0.0009	**0.0137****	0.0027	**0.0137****	0.0035	**0.0134****	0.0034
	(0.0020)	(0.0014)	(0.0030)	(0.0014)	(0.0028)	(0.0014)	(0.0057)	(0.0025)	(0.0062)	(0.0027)	(0.0061)	(0.0027)
Lagged_DV	**0.8558*****	**0.6146*****	**0.8612*****	**0.6322*****	**0.8617*****	**0.6337*****	**0.7214*****	**0.2209*****	**0.7227*****	**0.2377*****	**0.7231*****	**0.2385*****
	(0.0076)	(0.0133)	(0.0075)	(0.0130)	(0.0075)	(0.0130)	(0.0138)	(0.0201)	(0.0133)	(0.0201)	(0.0133)	(0.0201)
SalesGrowth			**-0.0178*****	**-0.0274*****	**-0.0169*****	**-0.0266*****			**-0.0156****	**-0.0171*****	**-0.0144****	**-0.0163*****
			(0.0045)	(0.0052)	(0.0044)	(0.0052)			(0.0061)	(0.0050)	(0.0060)	(0.0050)
RDSales			**0.0062*****	**-0.0032*****	**0.0062*****	**-0.0031****			**0.0137*****	-0.0014	**0.0138*****	-0.0012
			(0.0017)	(0.0012)	(0.0017)	(0.0012)			(0.0024)	(0.0019)	(0.0024)	(0.0019)
CashFlowAt			**0.1292*****	**0.1589*****	**0.1278*****	**0.1576*****			**0.1982*****	**0.1687*****	**0.1944*****	**0.1660*****
			(0.0162)	(0.0173)	(0.0162)	(0.0173)			(0.0236)	(0.0249)	(0.0237)	(0.0250)
DailyVola			0.0000	**-0.0001*****	**0.0001*****	**-0.0001***			**0.0003*****	**0.0001*****	**0.0003*****	0.0001
			(0.0000)	(0.0000)	(0.0000)	(0.0000)			(0.0000)	(0.0000)	(0.0001)	(0.0001)
Extended Controls			Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes
Industry FE	Yes		Yes		Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes		Yes		Yes
Year FE	Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes		
Year-Quarter FE					Yes	Yes					Yes	Yes
N	43,333	43,333	43,316	43,316	43,316	43,316	41,122	41,122	41,108	41,108	41,108	41,108
R^2	0.825	0.431	0.829	0.449	0.830	0.449	0.659	0.092	0.667	0.108	0.668	0.108
Adj. R^2	0.825	0.412	0.829	0.430	0.830	0.430	0.658	0.062	0.667	0.078	0.667	0.077

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). R^2 includes absorbed fixed effects (not within-R^2).

Table 9: H1 CEO 3-IV Decomp: ClarityCEO + UncResCEO (DWZ Qtr-Exp, no-look-ahead) + UncPreCEO

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	CashRatio						CashRatio_lead					
CEO Clarity (DWZ Qtr-Exp)	**-0.0083*****	**-0.0148*****	**-0.0093*****	**-0.0129*****	**-0.0092*****	**-0.0126*****	-0.0024	**-0.0215****	-0.0041	**-0.0210****	-0.0042	**-0.0210****
	(0.0033)	(0.0047)	(0.0033)	(0.0045)	(0.0033)	(0.0045)	(0.0053)	(0.0097)	(0.0053)	(0.0094)	(0.0053)	(0.0095)
CEO Residual Unc. (DWZ Qtr-Exp)	**0.0018***	**0.0020****	**0.0016***	**0.0018***	**0.0016***	**0.0017***	**0.0046*****	**0.0036****	**0.0044****	**0.0033****	**0.0044****	**0.0033****
	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0019)	(0.0017)	(0.0019)	(0.0017)	(0.0019)	(0.0017)
CEO Pres Uncertainty	0.0001	0.0006	-0.0001	0.0000	-0.0001	0.0001	0.0002	0.0016	-0.0003	0.0011	-0.0004	0.0011
	(0.0013)	(0.0013)	(0.0013)	(0.0013)	(0.0013)	(0.0013)	(0.0024)	(0.0021)	(0.0024)	(0.0021)	(0.0024)	(0.0021)
Leverage	**-0.0131*****	**-0.0324*****	**-0.0089*****	**-0.0199****	**-0.0087*****	**-0.0190****	**-0.0293****	**-0.0295****	**-0.0241****	-0.0181	**-0.0244****	-0.0179
	(0.0040)	(0.0086)	(0.0031)	(0.0085)	(0.0031)	(0.0085)	(0.0122)	(0.0139)	(0.0108)	(0.0136)	(0.0109)	(0.0136)
lnAssets	-0.0009	**-0.0169*****	**-0.0011***	**-0.0169*****	**-0.0011***	**-0.0167*****	**-0.0018***	**-0.0285*****	-0.0016	**-0.0286*****	-0.0015	**-0.0287*****
	(0.0006)	(0.0028)	(0.0006)	(0.0027)	(0.0006)	(0.0027)	(0.0011)	(0.0049)	(0.0010)	(0.0049)	(0.0011)	(0.0050)
TobinsQ	**0.0052*****	**0.0039*****	**0.0034*****	**0.0028****	**0.0033*****	**0.0025****	**0.0079*****	0.0011	**0.0046*****	-0.0009	**0.0046*****	-0.0009
	(0.0007)	(0.0011)	(0.0009)	(0.0012)	(0.0009)	(0.0012)	(0.0012)	(0.0017)	(0.0014)	(0.0017)	(0.0014)	(0.0018)
ROA	-0.0107	**0.0230***	**-0.0510*****	-0.0135	**-0.0498*****	-0.0129	**-0.0621*****	-0.0188	**-0.1280*****	**-0.0721*****	**-0.1270*****	**-0.0731*****
	(0.0109)	(0.0124)	(0.0131)	(0.0144)	(0.0132)	(0.0144)	(0.0192)	(0.0227)	(0.0202)	(0.0237)	(0.0204)	(0.0240)
Capex	**-0.0967*****	**-0.2472*****	**-0.1415*****	**-0.2628*****	**-0.1414*****	**-0.2605*****	**-0.1613*****	**-0.2816*****	**-0.2420*****	**-0.3002*****	**-0.2423*****	**-0.2996*****
	(0.0150)	(0.0253)	(0.0169)	(0.0258)	(0.0168)	(0.0257)	(0.0214)	(0.0372)	(0.0249)	(0.0375)	(0.0249)	(0.0375)
DivDummy	**-0.0037****	-0.0021	**-0.0051*****	-0.0034	**-0.0050*****	-0.0033	**-0.0042***	-0.0080	**-0.0048****	**-0.0090***	**-0.0047***	**-0.0091***
	(0.0015)	(0.0026)	(0.0015)	(0.0025)	(0.0015)	(0.0025)	(0.0025)	(0.0054)	(0.0025)	(0.0053)	(0.0025)	(0.0053)
sCFO	**0.0090*****	-0.0008	**0.0100*****	0.0004	**0.0099*****	0.0004	**0.0147***	0.0031	**0.0143***	0.0038	**0.0142***	0.0037
	(0.0024)	(0.0017)	(0.0031)	(0.0014)	(0.0030)	(0.0015)	(0.0081)	(0.0032)	(0.0082)	(0.0032)	(0.0081)	(0.0032)
Lagged_DV	**0.8718*****	**0.6433*****	**0.8734*****	**0.6576*****	**0.8738*****	**0.6579*****	**0.7424*****	**0.2158*****	**0.7404*****	**0.2320*****	**0.7402*****	**0.2317*****
	(0.0091)	(0.0159)	(0.0091)	(0.0156)	(0.0091)	(0.0156)	(0.0150)	(0.0242)	(0.0148)	(0.0242)	(0.0148)	(0.0242)
SalesGrowth			**-0.0165*****	**-0.0257*****	**-0.0165*****	**-0.0256*****			-0.0106	**-0.0137****	-0.0107	**-0.0136****
			(0.0054)	(0.0068)	(0.0054)	(0.0068)			(0.0070)	(0.0064)	(0.0070)	(0.0064)
RDSales			**0.0091*****	-0.0004	**0.0091*****	-0.0003			**0.0160*****	0.0005	**0.0159*****	0.0005
			(0.0025)	(0.0017)	(0.0025)	(0.0017)			(0.0038)	(0.0021)	(0.0038)	(0.0021)
CashFlowAt			**0.1182*****	**0.1536*****	**0.1189*****	**0.1544*****			**0.1977*****	**0.1883*****	**0.1974*****	**0.1880*****
			(0.0159)	(0.0175)	(0.0159)	(0.0175)			(0.0275)	(0.0300)	(0.0275)	(0.0300)
DailyVola			0.0000	**-0.0001****	0.0000	**-0.0001****			**0.0002*****	0.0000	**0.0003*****	0.0000
			(0.0000)	(0.0000)	(0.0000)	(0.0000)			(0.0001)	(0.0000)	(0.0001)	(0.0001)
Extended Controls			Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes
Industry FE	Yes		Yes		Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes		Yes		Yes
Year FE	Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes		
Year-Quarter FE					Yes	Yes					Yes	Yes
N	24,868	24,868	24,855	24,855	24,855	24,855	23,127	23,127	23,120	23,120	23,120	23,120
R^2	0.839	0.457	0.842	0.473	0.842	0.473	0.675	0.087	0.683	0.105	0.683	0.105
Adj. R^2	0.839	0.430	0.841	0.446	0.842	0.446	0.674	0.041	0.682	0.059	0.682	0.058

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). R^2 includes absorbed fixed effects (not within-R^2).

Table 10: H1.2 CEO 3-IV Decomp HFC: ClarityCEO + UncResCEO (DWZ Full) + UncPreCEO × Unrated Constraint

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	CashRatio				CashRatio_lead			
CEO Clarity (DWZ, c)	**-0.0062****	-0.0043	**-0.0059****	-0.0042	-0.0052	**-0.0157***	-0.0052	**-0.0155***
	(0.0034)	(0.0052)	(0.0034)	(0.0052)	(0.0053)	(0.0096)	(0.0053)	(0.0096)
CEO Residual Unc. (DWZ, c)	**0.0020****	**0.0018****	**0.0022****	**0.0019****	**0.0027****	**0.0022***	**0.0026****	**0.0021***
	(0.0011)	(0.0010)	(0.0011)	(0.0010)	(0.0015)	(0.0013)	(0.0015)	(0.0013)
CEO Pres Unc. (c)	0.0002	0.0005	0.0003	0.0007	-0.0007	-0.0001	-0.0008	-0.0002
	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0022)	(0.0018)	(0.0022)	(0.0018)
Unrated	**0.0038*****	-0.0027	-0.0010	-0.0026	**0.0065*****	**-0.0093***	-0.0014	**-0.0095***
	(0.0012)	(0.0035)	(0.0016)	(0.0035)	(0.0022)	(0.0054)	(0.0029)	(0.0054)
UncRes × Unrated	0.0011	-0.0001	0.0012	-0.0000	**0.0057****	0.0033	**0.0057****	0.0033
	(0.0021)	(0.0020)	(0.0021)	(0.0020)	(0.0029)	(0.0026)	(0.0029)	(0.0026)
UncPre × Unrated	0.0024	0.0013	**0.0029***	0.0014	0.0005	-0.0026	0.0010	-0.0027
	(0.0022)	(0.0025)	(0.0023)	(0.0025)	(0.0040)	(0.0038)	(0.0040)	(0.0038)
Leverage	**-0.0117*****	**-0.0216****	**-0.0141*****	**-0.0214****	**-0.0237****	-0.0107	**-0.0274****	-0.0104
	(0.0033)	(0.0098)	(0.0041)	(0.0099)	(0.0103)	(0.0131)	(0.0117)	(0.0131)
lnAssets	**0.0005***	**-0.0159*****	**-0.0019*****	**-0.0155*****	**0.0018*****	**-0.0273*****	**-0.0025****	**-0.0275*****
	(0.0003)	(0.0027)	(0.0007)	(0.0027)	(0.0005)	(0.0044)	(0.0011)	(0.0044)
TobinsQ	**0.0047*****	**0.0045*****	**0.0041*****	**0.0040*****	**0.0059*****	0.0009	**0.0052*****	0.0010
	(0.0008)	(0.0011)	(0.0008)	(0.0011)	(0.0013)	(0.0017)	(0.0013)	(0.0017)
ROA	**-0.0715*****	-0.0157	**-0.0651*****	-0.0133	**-0.1452*****	**-0.0819*****	**-0.1378*****	**-0.0800*****
	(0.0122)	(0.0122)	(0.0122)	(0.0122)	(0.0182)	(0.0200)	(0.0183)	(0.0202)
Capex	**-0.1610*****	**-0.3087*****	**-0.1647*****	**-0.3005*****	**-0.2338*****	**-0.3399*****	**-0.2426*****	**-0.3419*****
	(0.0158)	(0.0254)	(0.0158)	(0.0251)	(0.0226)	(0.0360)	(0.0227)	(0.0361)
DivDummy	**-0.0058*****	**-0.0043***	**-0.0054*****	-0.0038	**-0.0054****	-0.0065	**-0.0055****	-0.0066
	(0.0013)	(0.0023)	(0.0013)	(0.0023)	(0.0023)	(0.0044)	(0.0023)	(0.0044)
sCFO	**0.0081*****	-0.0003	**0.0072*****	-0.0005	**0.0135****	0.0043	**0.0125****	0.0042
	(0.0031)	(0.0016)	(0.0026)	(0.0016)	(0.0062)	(0.0029)	(0.0055)	(0.0029)
SalesGrowth	**-0.0189*****	**-0.0297*****	**-0.0176*****	**-0.0287*****	**-0.0158****	**-0.0156*****	**-0.0143****	**-0.0148*****
	(0.0045)	(0.0048)	(0.0044)	(0.0048)	(0.0062)	(0.0046)	(0.0062)	(0.0047)
RDSales	**0.0089*****	-0.0018	**0.0090*****	-0.0016	**0.0152*****	-0.0004	**0.0155*****	-0.0003
	(0.0020)	(0.0015)	(0.0020)	(0.0015)	(0.0032)	(0.0024)	(0.0032)	(0.0024)
CashFlowAt	**0.1415*****	**0.1741*****	**0.1368*****	**0.1729*****	**0.2109*****	**0.1796*****	**0.2012*****	**0.1767*****
	(0.0166)	(0.0180)	(0.0165)	(0.0180)	(0.0240)	(0.0257)	(0.0241)	(0.0259)
DailyVola	**0.0001****	**-0.0001*****	**0.0001****	-0.0001	**0.0004*****	**0.0001*****	**0.0003*****	0.0001
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0000)	(0.0001)	(0.0001)
Lagged_DV	**0.8605*****	**0.6188*****	**0.8571*****	**0.6204*****	**0.7310*****	**0.2231*****	**0.7250*****	**0.2242*****
	(0.0079)	(0.0133)	(0.0080)	(0.0134)	(0.0135)	(0.0193)	(0.0138)	(0.0193)
Industry FE	Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes
Year FE	Yes	Yes			Yes	Yes		
Year-Quarter FE			Yes	Yes			Yes	Yes
N	38,737	38,737	38,737	38,737	38,092	38,092	38,092	38,092
R^2	0.827	0.432	0.829	0.432	0.667	0.100	0.669	0.100
Adj. R^2	0.827	0.412	0.828	0.411	0.667	0.069	0.668	0.067

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Fiscal years 2002-2016. DWZ Eq.5 decomposition: Clarity = -CEO FE; UncRes = call-level residual. R^2 includes absorbed fixed effects (not within-R^2).

Table 11: H1.2 CEO 3-IV Decomp HFC: ClarityCEO + UncResCEO (DWZ Qtr-Exp, no-look-ahead) + UncPreCEO × Unrated Constraint

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	CashRatio				CashRatio_lead			
CEO Clarity (DWZ Qtr-Exp, c)	**-0.0114*****	**-0.0132*****	**-0.0104*****	**-0.0127*****	-0.0037	**-0.0186****	-0.0019	**-0.0187****
	(0.0035)	(0.0052)	(0.0034)	(0.0053)	(0.0054)	(0.0103)	(0.0054)	(0.0103)
CEO Residual Unc. (DWZ Qtr-Exp, c)	**0.0021***	**0.0024****	**0.0021***	**0.0022****	**0.0044****	**0.0036****	**0.0044****	**0.0036****
	(0.0013)	(0.0013)	(0.0013)	(0.0013)	(0.0020)	(0.0017)	(0.0020)	(0.0017)
CEO Pres Unc. (c)	0.0000	-0.0005	0.0000	-0.0004	0.0003	0.0009	0.0000	0.0009
	(0.0013)	(0.0013)	(0.0013)	(0.0013)	(0.0025)	(0.0022)	(0.0025)	(0.0022)
Unrated	0.0021	-0.0043	-0.0012	-0.0043	**0.0051****	-0.0073	-0.0019	-0.0074
	(0.0014)	(0.0043)	(0.0018)	(0.0043)	(0.0024)	(0.0066)	(0.0032)	(0.0066)
UncRes (Qtr-Exp) × Unrated	0.0019	0.0013	0.0019	0.0013	**0.0050***	0.0023	**0.0050***	0.0022
	(0.0025)	(0.0023)	(0.0025)	(0.0023)	(0.0038)	(0.0033)	(0.0038)	(0.0033)
UncPre × Unrated	0.0014	-0.0014	0.0017	-0.0014	-0.0001	-0.0015	0.0004	-0.0014
	(0.0026)	(0.0029)	(0.0026)	(0.0029)	(0.0046)	(0.0046)	(0.0046)	(0.0046)
Leverage	**-0.0084*****	-0.0170	**-0.0094*****	-0.0160	**-0.0210***	-0.0110	**-0.0240****	-0.0107
	(0.0031)	(0.0104)	(0.0034)	(0.0104)	(0.0108)	(0.0154)	(0.0121)	(0.0154)
lnAssets	0.0003	**-0.0157*****	**-0.0016***	**-0.0157*****	**0.0016*****	**-0.0300*****	**-0.0021***	**-0.0302*****
	(0.0003)	(0.0031)	(0.0008)	(0.0031)	(0.0006)	(0.0052)	(0.0013)	(0.0053)
TobinsQ	**0.0039*****	**0.0030****	**0.0034*****	**0.0025***	**0.0049*****	-0.0011	**0.0044*****	-0.0011
	(0.0010)	(0.0015)	(0.0010)	(0.0014)	(0.0014)	(0.0019)	(0.0015)	(0.0020)
ROA	**-0.0554*****	-0.0071	**-0.0535*****	-0.0065	**-0.1336*****	**-0.0761*****	**-0.1305*****	**-0.0773*****
	(0.0139)	(0.0145)	(0.0140)	(0.0146)	(0.0211)	(0.0249)	(0.0213)	(0.0252)
Capex	**-0.1396*****	**-0.2564*****	**-0.1439*****	**-0.2536*****	**-0.2276*****	**-0.2870*****	**-0.2378*****	**-0.2865*****
	(0.0175)	(0.0276)	(0.0176)	(0.0275)	(0.0263)	(0.0384)	(0.0262)	(0.0384)
DivDummy	**-0.0055*****	-0.0041	**-0.0056*****	-0.0041	**-0.0044***	-0.0076	**-0.0046***	-0.0077
	(0.0016)	(0.0027)	(0.0016)	(0.0026)	(0.0025)	(0.0053)	(0.0026)	(0.0053)
sCFO	**0.0099*****	0.0017	**0.0094*****	0.0017	**0.0123***	0.0041	**0.0116***	0.0040
	(0.0030)	(0.0016)	(0.0028)	(0.0016)	(0.0069)	(0.0032)	(0.0063)	(0.0032)
SalesGrowth	**-0.0128****	**-0.0267*****	**-0.0128****	**-0.0265*****	-0.0081	**-0.0127****	-0.0083	**-0.0127****
	(0.0051)	(0.0056)	(0.0051)	(0.0056)	(0.0074)	(0.0063)	(0.0075)	(0.0063)
RDSales	**0.0086*****	-0.0004	**0.0086*****	-0.0003	**0.0154*****	0.0010	**0.0154*****	0.0011
	(0.0022)	(0.0017)	(0.0022)	(0.0017)	(0.0035)	(0.0022)	(0.0035)	(0.0022)
CashFlowAt	**0.1220*****	**0.1745*****	**0.1204*****	**0.1754*****	**0.2021*****	**0.1992*****	**0.1968*****	**0.1990*****
	(0.0169)	(0.0188)	(0.0169)	(0.0189)	(0.0288)	(0.0317)	(0.0288)	(0.0317)
DailyVola	0.0000	**-0.0001****	-0.0000	**-0.0001****	**0.0003*****	0.0000	**0.0002*****	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0000)	(0.0001)	(0.0001)
Lagged_DV	**0.8737*****	**0.6448*****	**0.8713*****	**0.6451*****	**0.7471*****	**0.2124*****	**0.7408*****	**0.2122*****
	(0.0096)	(0.0166)	(0.0096)	(0.0166)	(0.0152)	(0.0230)	(0.0154)	(0.0229)
Industry FE	Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes
Year FE	Yes	Yes			Yes	Yes		
Year-Quarter FE			Yes	Yes			Yes	Yes
N	21,824	21,824	21,824	21,824	21,371	21,371	21,371	21,371
R^2	0.841	0.453	0.841	0.453	0.679	0.094	0.680	0.094
Adj. R^2	0.841	0.423	0.841	0.423	0.678	0.046	0.679	0.044

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Fiscal years 2002-2016. DWZ Eq.5 decomposition under quarterly-expanding window (no-look-ahead). R^2 includes absorbed fixed effects (not within-R^2).

Table 12: H1.3 CEO 3-IV Decomp CFvol: ClarityCEO + UncResCEO (DWZ Full) + UncPreCEO × HighCFvol (Han-Qiu 16q CV)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	CashRatio				CashRatio_lead			
CEO Clarity (DWZ, c)	**-0.0088*****	-0.0059	**-0.0071****	-0.0058	**-0.0096****	**-0.0135***	**-0.0078***	**-0.0134***
	(0.0034)	(0.0047)	(0.0033)	(0.0047)	(0.0052)	(0.0091)	(0.0052)	(0.0091)
CEO Residual Unc. (DWZ, c)	**0.0021****	**0.0017***	**0.0022****	**0.0017****	**0.0025****	0.0016	**0.0024***	0.0015
	(0.0011)	(0.0010)	(0.0011)	(0.0010)	(0.0015)	(0.0013)	(0.0015)	(0.0014)
CEO Pres Unc. (c)	0.0004	0.0006	0.0001	0.0007	-0.0002	0.0004	-0.0008	0.0002
	(0.0011)	(0.0011)	(0.0012)	(0.0012)	(0.0021)	(0.0018)	(0.0022)	(0.0018)
HighCFvol	-0.0005	-0.0022	**-0.0028****	-0.0023	**0.0075*****	0.0021	**0.0049****	0.0021
	(0.0012)	(0.0016)	(0.0013)	(0.0016)	(0.0019)	(0.0025)	(0.0020)	(0.0025)
UncRes × HighCFvol	0.0004	0.0004	-0.0000	0.0002	**0.0047***	0.0028	**0.0048***	0.0030
	(0.0020)	(0.0019)	(0.0020)	(0.0019)	(0.0030)	(0.0026)	(0.0029)	(0.0026)
UncPre × HighCFvol	**0.0048****	0.0002	**0.0049****	0.0002	**0.0099*****	0.0040	**0.0101*****	0.0041
	(0.0022)	(0.0024)	(0.0022)	(0.0024)	(0.0039)	(0.0040)	(0.0039)	(0.0040)
Leverage	**-0.0127*****	**-0.0191****	**-0.0130*****	**-0.0189****	**-0.0273*****	-0.0116	**-0.0274*****	-0.0112
	(0.0034)	(0.0082)	(0.0036)	(0.0082)	(0.0094)	(0.0121)	(0.0097)	(0.0121)
lnAssets	0.0004	**-0.0174*****	**-0.0015*****	**-0.0169*****	**0.0014*****	**-0.0236*****	-0.0013	**-0.0235*****
	(0.0003)	(0.0026)	(0.0006)	(0.0026)	(0.0005)	(0.0043)	(0.0010)	(0.0044)
TobinsQ	**0.0044*****	**0.0037*****	**0.0035*****	**0.0034*****	**0.0066*****	0.0016	**0.0056*****	0.0017
	(0.0008)	(0.0011)	(0.0008)	(0.0011)	(0.0013)	(0.0017)	(0.0014)	(0.0018)
ROA	**-0.0641*****	-0.0148	**-0.0563*****	-0.0121	**-0.1405*****	**-0.0736*****	**-0.1330*****	**-0.0717*****
	(0.0117)	(0.0123)	(0.0116)	(0.0123)	(0.0185)	(0.0205)	(0.0186)	(0.0206)
Capex	**-0.1692*****	**-0.3164*****	**-0.1779*****	**-0.3085*****	**-0.2443*****	**-0.3482*****	**-0.2565*****	**-0.3496*****
	(0.0158)	(0.0251)	(0.0160)	(0.0249)	(0.0229)	(0.0363)	(0.0232)	(0.0364)
DivDummy	**-0.0057*****	**-0.0051****	**-0.0054*****	**-0.0046****	**-0.0048****	**-0.0085***	**-0.0049****	**-0.0086***
	(0.0013)	(0.0023)	(0.0013)	(0.0023)	(0.0023)	(0.0046)	(0.0023)	(0.0046)
sCFO	**0.0087****	0.0004	**0.0078****	0.0003	**0.0128***	0.0041	**0.0121***	0.0040
	(0.0043)	(0.0017)	(0.0037)	(0.0017)	(0.0071)	(0.0031)	(0.0066)	(0.0030)
SalesGrowth	**-0.0203*****	**-0.0291*****	**-0.0192*****	**-0.0283*****	**-0.0187*****	**-0.0188*****	**-0.0172****	**-0.0181*****
	(0.0050)	(0.0057)	(0.0049)	(0.0057)	(0.0068)	(0.0055)	(0.0067)	(0.0056)
RDSales	**0.0063*****	**-0.0031****	**0.0063*****	**-0.0029****	**0.0153*****	-0.0015	**0.0154*****	-0.0014
	(0.0022)	(0.0015)	(0.0022)	(0.0015)	(0.0037)	(0.0029)	(0.0036)	(0.0029)
CashFlowAt	**0.1476*****	**0.1713*****	**0.1382*****	**0.1698*****	**0.2228*****	**0.1725*****	**0.2093*****	**0.1698*****
	(0.0170)	(0.0179)	(0.0171)	(0.0179)	(0.0255)	(0.0262)	(0.0258)	(0.0264)
DailyVola	**0.0001*****	**-0.0001*****	**0.0001*****	-0.0001	**0.0004*****	**0.0001*****	**0.0003*****	0.0001
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0000)	(0.0001)	(0.0001)
Lagged_DV	**0.8664*****	**0.6307*****	**0.8619*****	**0.6322*****	**0.7262*****	**0.2312*****	**0.7200*****	**0.2320*****
	(0.0077)	(0.0133)	(0.0078)	(0.0133)	(0.0139)	(0.0205)	(0.0142)	(0.0205)
Industry FE	Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes
Year FE	Yes	Yes			Yes	Yes		
Year-Quarter FE			Yes	Yes			Yes	Yes
N	40,795	40,795	40,795	40,795	38,671	38,671	38,671	38,671
R^2	0.825	0.447	0.826	0.447	0.657	0.103	0.659	0.102
Adj. R^2	0.824	0.429	0.826	0.428	0.657	0.072	0.658	0.071

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Fiscal years 2002-2018. Full panel coverage via Compustat Quarterly OCF Extended (1990-2002 backfill); Han-Qiu 16-trailing-quarter CFvol available from 1994-Q1 onwards in source data. R^2 includes absorbed fixed effects (not within-R^2).

Table 13: H11: Political Risk and Language Uncertainty

	(1)	(2)	(3)	(4)
	Industry FE		Firm FE	
	UncResCEO	UncPreCEO	UncResCEO	UncPreCEO
Political Risk$_t$	**0.0001*****	**0.0003*****	**0.0001*****	**0.0002*****
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
UncQue	-0.0022	**0.0204*****	-0.0014	**0.0081*****
	(0.0032)	(0.0045)	(0.0037)	(0.0029)
NegCall	-0.0025	**0.1874*****	-0.0036	**0.1477*****
	(0.0051)	(0.0141)	(0.0073)	(0.0082)
lnAssets	-0.0002	**-0.0296*****	-0.0054	0.0048
	(0.0004)	(0.0039)	(0.0058)	(0.0078)
TobinsQ	-0.0005	-0.0007	-0.0008	0.0001
	(0.0009)	(0.0036)	(0.0022)	(0.0023)
ROA	**0.0291****	0.0477	**0.0540****	0.0346
	(0.0145)	(0.0318)	(0.0264)	(0.0230)
CashRatio	0.0019	-0.0354	0.0206	0.0265
	(0.0077)	(0.0357)	(0.0240)	(0.0287)
DivDummy	-0.0031	-0.0139	-0.0108	**-0.0229****
	(0.0021)	(0.0118)	(0.0074)	(0.0103)
FirmMat	0.0012	**0.0021****	0.0027	0.0006
	(0.0013)	(0.0010)	(0.0041)	(0.0007)
EarnVol	0.0078	-0.0072	0.0056	-0.0104
	(0.0077)	(0.0157)	(0.0117)	(0.0132)
UncPreCEO	-0.0051		-0.0054	
	(0.0034)		(0.0054)	
Industry FE	Yes	Yes		
Firm FE			Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	44,497	65,760	44,497	65,760
R^2	0.003	0.043	0.003	0.023
Adj. R^2	0.002	0.043	-0.030	-0.004

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Firms with fewer than 5 calls are excluded. R^2 includes absorbed fixed effects (not within-R^2).

Table 14: H11-Lag2: Political Risk (2-Qtr Lag) and Language Uncertainty

	(1)	(2)	(3)	(4)
	Industry FE		Firm FE	
	UncResCEO	UncPreCEO	UncResCEO	UncPreCEO
PRisk_{t-2}	0.0000	**0.0001*****	0.0000	**0.0000*****
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
UncQue	0.0001	**0.0242*****	0.0010	**0.0099*****
	(0.0033)	(0.0047)	(0.0038)	(0.0031)
NegCall	0.0020	**0.2019*****	0.0019	**0.1609*****
	(0.0052)	(0.0145)	(0.0074)	(0.0084)
lnAssets	-0.0001	**-0.0298*****	-0.0035	0.0047
	(0.0005)	(0.0040)	(0.0062)	(0.0078)
TobinsQ	-0.0005	-0.0002	-0.0011	-0.0002
	(0.0010)	(0.0037)	(0.0023)	(0.0023)
ROA	**0.0293***	0.0518	**0.0598****	**0.0464***
	(0.0152)	(0.0331)	(0.0272)	(0.0247)
CashRatio	0.0064	-0.0400	0.0170	0.0233
	(0.0076)	(0.0368)	(0.0242)	(0.0292)
DivDummy	-0.0031	-0.0142	**-0.0133***	**-0.0207***
	(0.0022)	(0.0121)	(0.0076)	(0.0106)
FirmMat	0.0013	0.0021	0.0025	-0.0007
	(0.0013)	(0.0015)	(0.0041)	(0.0016)
EarnVol	0.0096	-0.0078	0.0089	-0.0154
	(0.0063)	(0.0159)	(0.0100)	(0.0158)
UncPreCEO	-0.0001		-0.0003	
	(0.0035)		(0.0056)	
Industry FE	Yes	Yes		
Firm FE			Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	43,037	63,022	43,037	63,022
R^2	0.000	0.035	0.000	0.015
Adj. R^2	-0.001	0.034	-0.033	-0.013

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Firms with fewer than 5 calls are excluded. R^2 includes absorbed fixed effects (not within-R^2).

Table 15: H23: Product-Market Competition and Uncertainty Language

	(1)	(2)	(3)	(4)
	Industry FE		Firm FE	
	UncResCEO	UncPreCEO	UncResCEO	UncPreCEO
$z(\log(\text{TSIMM}))$	0.0008	**0.0304*****	0.0023	**0.0302*****
	(0.0014)	(0.0074)	(0.0075)	(0.0093)
UncQue	0.0011	**0.0398*****	0.0039	0.0084
	(0.0062)	(0.0100)	(0.0087)	(0.0073)
NegCall	0.0020	**0.2340*****	0.0046	**0.1698*****
	(0.0068)	(0.0201)	(0.0120)	(0.0140)
lnAssets	-0.0007	**-0.0335*****	-0.0101	0.0013
	(0.0007)	(0.0038)	(0.0071)	(0.0081)
TobinsQ	0.0001	-0.0007	0.0009	-0.0017
	(0.0013)	(0.0037)	(0.0025)	(0.0025)
ROA	0.0233	**0.0681****	0.0343	0.0241
	(0.0197)	(0.0314)	(0.0308)	(0.0254)
CashRatio	0.0065	**-0.0803****	0.0055	0.0022
	(0.0103)	(0.0360)	(0.0287)	(0.0306)
DivDummy	-0.0020	-0.0060	-0.0074	-0.0166
	(0.0028)	(0.0117)	(0.0083)	(0.0106)
FirmMat	0.0018	**0.0014****	0.0083	0.0003
	(0.0019)	(0.0007)	(0.0056)	(0.0004)
EarnVol	-0.0047	-0.0210	-0.0048	-0.0114
	(0.0087)	(0.0152)	(0.0121)	(0.0146)
UncPreCEO	**0.0111****		**0.0230****	
	(0.0044)		(0.0093)	
Industry FE	Yes	Yes		
Firm FE			Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	12,728	18,492	12,728	18,492
R^2	0.001	0.050	0.001	0.021
Adj. R^2	-0.002	0.048	-0.098	-0.068

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Unit of observation: firm-fiscal-year. R^2 includes absorbed fixed effects (not within-R^2).

Table 16: H24: US Economic Policy Uncertainty and Call Language Uncertainty

	(1) Industry + Cal. Year FE	(2) Industry + Cal. Year FE	(3) Firm + Cal. Year FE	(4) Firm + Cal. Year FE
	UncResCEO	UncPreCEO	UncResCEO	UncPreCEO
$\log(\text{US EPU})_t$	**0.0124****	**0.0211*****	**0.0123***	**0.0239*****
	(0.0075)	(0.0074)	(0.0075)	(0.0076)
UncQue	0.0011	**0.0164*****	0.0024	**0.0095*****
	(0.0035)	(0.0032)	(0.0040)	(0.0028)
NegCall	0.0006	**0.1247*****	0.0021	**0.1385*****
	(0.0056)	(0.0089)	(0.0073)	(0.0079)
lnAssets	0.0009	**-0.0143*****	0.0013	0.0038
	(0.0007)	(0.0020)	(0.0067)	(0.0063)
TobinsQ	-0.0001	0.0007	-0.0000	0.0007
	(0.0011)	(0.0019)	(0.0027)	(0.0018)
ROA	**0.0281***	**0.0367****	**0.0677****	**0.0467****
	(0.0149)	(0.0168)	(0.0268)	(0.0191)
CashRatio	0.0051	-0.0079	0.0345	0.0305
	(0.0087)	(0.0187)	(0.0256)	(0.0231)
DivDummy	**-0.0046*****	-0.0074	**-0.0142****	**-0.0157***
	(0.0009)	(0.0061)	(0.0072)	(0.0083)
FirmMat	**0.0023***	0.0009	0.0005	-0.0011
	(0.0012)	(0.0008)	(0.0047)	(0.0012)
EarnVol	**0.0268***	-0.0074	**0.0353****	-0.0083
	(0.0157)	(0.0086)	(0.0176)	(0.0119)
UncPreCEO	-0.0026		-0.0011	
	(0.0032)		(0.0045)	
Lagged_DV	**0.0202*****	**0.5176*****	**0.0187*****	**0.2626*****
	(0.0054)	(0.0131)	(0.0056)	(0.0089)
Industry FE	Yes	Yes		
Firm FE			Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	39,063	60,503	39,063	60,503
R^2	0.001	0.292	0.001	0.084
Adj. R^2	-0.000	0.291	-0.035	0.057

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) two-way clustered (firm, calendar quarter). Main sample (excludes financial and utility firms). Firms with fewer than 5 calls are excluded. R^2 includes absorbed fixed effects (not within-R^2).

Table 17: H24b: Global Economic Policy Uncertainty and Call Language Uncertainty

	(1)	(2)	(3)	(4)
	Industry + Cal. Year FE		Firm + Cal. Year FE	
	UncResCEO	UncPreCEO	UncResCEO	UncPreCEO
$\log(\text{GEPU})_t$	**0.0181****	**0.0271*****	**0.0187****	**0.0309*****
	(0.0100)	(0.0107)	(0.0101)	(0.0108)
UncQue	0.0011	**0.0164*****	0.0024	**0.0095*****
	(0.0035)	(0.0032)	(0.0039)	(0.0028)
NegCall	0.0006	**0.1247*****	0.0020	**0.1385*****
	(0.0056)	(0.0089)	(0.0072)	(0.0079)
lnAssets	0.0009	**-0.0143*****	0.0012	0.0037
	(0.0007)	(0.0020)	(0.0067)	(0.0063)
TobinsQ	-0.0001	0.0008	0.0000	0.0007
	(0.0011)	(0.0019)	(0.0027)	(0.0018)
ROA	**0.0280***	**0.0367****	**0.0673****	**0.0462****
	(0.0149)	(0.0168)	(0.0267)	(0.0191)
CashRatio	0.0051	-0.0080	0.0345	0.0304
	(0.0087)	(0.0188)	(0.0256)	(0.0231)
DivDummy	**-0.0047*****	-0.0074	**-0.0142****	**-0.0156***
	(0.0009)	(0.0061)	(0.0072)	(0.0083)
FirmMat	**0.0023***	0.0009	0.0006	-0.0011
	(0.0012)	(0.0008)	(0.0047)	(0.0012)
EarnVol	**0.0269***	-0.0074	**0.0354****	-0.0082
	(0.0157)	(0.0086)	(0.0176)	(0.0119)
UncPreCEO	-0.0026		-0.0011	
	(0.0032)		(0.0045)	
Lagged_DV	**0.0202*****	**0.5176*****	**0.0187*****	**0.2626*****
	(0.0054)	(0.0131)	(0.0056)	(0.0089)
Industry FE	Yes	Yes		
Firm FE			Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	39,063	60,503	39,063	60,503
R^2	0.001	0.292	0.001	0.084
Adj. R^2	-0.000	0.291	-0.035	0.057

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) two-way clustered (firm, calendar quarter). Main sample (excludes financial and utility firms). Firms with fewer than 5 calls are excluded. R^2 includes absorbed fixed effects (not within-R^2).

Table 18: H14c CEO 3-IV Decomp: ClarityCEO + UncResCEO (DWZ Eq.5 Full) + UncPreCEO (raw) and 25-Day Post-Call Bid-Ask Spread Level ($[0, +25]$, the call day plus 25 trading days)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Spread$_{25D,t}$						Spread$_{25D,t+1}$					
CEO Clarity (DWZ)	-0.0477	1.0617	0.2122	1.1056	0.1644	1.0547	-0.1886	0.5998	-0.0922	0.6828	-0.0431	0.5722
	(0.2145)	(0.4277)	(0.2227)	(0.4260)	(0.2103)	(0.4174)	(0.2370)	(0.4004)	(0.2399)	(0.4055)	(0.2206)	(0.3729)
CEO Residual Uncertainty (DWZ)	-0.0594	-0.0687	-0.0882	-0.1021	-0.0757	-0.0893	0.0825	0.0696	0.0742	0.0545	0.0984	0.0751
	(0.1068)	(0.1057)	(0.1023)	(0.1007)	(0.0994)	(0.0981)	(0.1416)	(0.1437)	(0.1407)	(0.1426)	(0.1339)	(0.1357)
CEO Pres Uncertainty	**0.1664****	**0.3496*****	0.0814	**0.2945*****	0.0108	**0.1644***	-0.0817	-0.0288	-0.1072	-0.0360	0.0170	0.1124
	(0.0946)	(0.1253)	(0.0956)	(0.1229)	(0.0919)	(0.1184)	(0.0926)	(0.1229)	(0.0926)	(0.1233)	(0.0872)	(0.1186)
lnAssets	**-0.8039*****	**-1.9098*****	**-0.6339*****	**-1.6825*****	**-0.6352*****	**-1.8277*****	**-0.8218*****	**-1.5191*****	**-0.7857*****	**-1.7292*****	**-0.6568*****	**-1.4181*****
	(0.0490)	(0.2000)	(0.0472)	(0.2156)	(0.0473)	(0.2158)	(0.0513)	(0.1951)	(0.0505)	(0.2130)	(0.0474)	(0.2017)
TobinsQ	**-0.2861*****	**-0.4785*****	**-0.2633*****	**-0.5518*****	**-0.2654*****	**-0.5883*****	**-0.2305*****	**-0.3215*****	**-0.2285*****	**-0.4803*****	**-0.2035*****	**-0.4289*****
	(0.0457)	(0.0900)	(0.0470)	(0.0963)	(0.0451)	(0.0956)	(0.0467)	(0.0850)	(0.0478)	(0.1002)	(0.0433)	(0.0935)
ROA	**-9.4469*****	**-11.6430*****	**-5.7059*****	**-7.6844*****	**-5.7511*****	**-7.8517*****	**-9.9256*****	**-12.8443*****	**-8.8744*****	**-11.9197*****	**-7.6375*****	**-10.5867*****
	(1.0201)	(1.3867)	(0.9822)	(1.2654)	(0.9633)	(1.2490)	(1.0798)	(1.2992)	(1.0722)	(1.2732)	(0.9869)	(1.1921)
Leverage	**0.3875***	**1.8330*****	-0.1126	0.6850	-0.0542	0.8026	0.2924	0.5282	0.1338	0.2221	0.0592	0.1928
	(0.2088)	(0.5143)	(0.2434)	(0.5013)	(0.2264)	(0.4954)	(0.2224)	(0.5516)	(0.2248)	(0.5596)	(0.2111)	(0.5258)
Capex	-0.8001	0.9540	**-2.3622****	-1.5271	**-2.0790***	-1.2897	**-1.6463***	-3.2121	**-2.1561****	**-4.3379***	-1.4770	-2.4356
	(1.1109)	(2.9126)	(1.1044)	(2.8061)	(1.0845)	(2.7998)	(0.9620)	(2.2020)	(0.9823)	(2.2446)	(0.9011)	(2.1565)
DivDummy	0.1043	0.1429	**0.3329*****	0.2549	**0.3156*****	0.2723	0.0240	0.1007	0.1012	0.1454	0.1145	0.1739
	(0.0893)	(0.2020)	(0.0896)	(0.1888)	(0.0855)	(0.1867)	(0.0887)	(0.2095)	(0.0889)	(0.2074)	(0.0825)	(0.1975)
sCFO	**0.6700****	**0.8405****	0.4045	**0.6825***	0.4084	**0.6882****	0.0871	**0.3644***	0.0040	0.3165	-0.0840	0.2479
	(0.3405)	(0.3472)	(0.4604)	(0.3789)	(0.4126)	(0.3375)	(0.1759)	(0.1877)	(0.2174)	(0.2055)	(0.1937)	(0.1738)
Lagged_DV	**0.7547*****	**0.6569*****	**0.7140*****	**0.6180*****	**0.7310*****	**0.6318*****	**0.7540*****	**0.6361*****	**0.7360*****	**0.6135*****	**0.7676*****	**0.6518*****
	(0.0150)	(0.0186)	(0.0167)	(0.0199)	(0.0168)	(0.0208)	(0.0156)	(0.0185)	(0.0173)	(0.0205)	(0.0162)	(0.0197)
StockPrice			-0.0011	-0.0033	0.0002	0.0019			-0.0001	**0.0101*****	-0.0016	**0.0047****
			(0.0014)	(0.0024)	(0.0013)	(0.0024)			(0.0013)	(0.0024)	(0.0012)	(0.0022)
Turnover			**-24.1012*****	**-23.8647*****	**-20.7840*****	**-20.3411*****			**-7.9396*****	**-5.5050*****	**-7.1303*****	**-4.7073****
			(1.7508)	(2.2119)	(1.7290)	(2.1791)			(1.5411)	(2.0904)	(1.4699)	(1.9633)
DailyVola			**0.1229*****	**0.1393*****	**0.1075*****	**0.1257*****			**0.0404*****	**0.0520*****	**0.0396*****	**0.0441*****
			(0.0072)	(0.0081)	(0.0082)	(0.0099)			(0.0062)	(0.0068)	(0.0069)	(0.0081)
AbsSurpDec			**-0.0919*****	-0.0241	**-0.0756*****	-0.0093			-0.0050	0.0119	-0.0174	-0.0012
			(0.0265)	(0.0300)	(0.0253)	(0.0286)			(0.0278)	(0.0304)	(0.0261)	(0.0285)
Extended Controls			Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes
Industry FE	Yes		Yes		Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes		Yes		Yes
Year FE	Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes		
Year-Quarter FE					Yes	Yes					Yes	Yes
N	42,625	42,625	42,625	42,625	42,625	42,625	43,049	43,049	43,049	43,049	43,049	43,049
R^2	0.703	0.517	0.721	0.551	0.730	0.548	0.700	0.499	0.702	0.504	0.723	0.523
Adj. R^2	0.703	0.501	0.721	0.535	0.730	0.532	0.700	0.482	0.702	0.487	0.723	0.506

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). DWZ Full decomposition: Clarity = -CEO FE; UncRes = call-level residual. R^2 includes absorbed fixed effects (not within-R^2). Spread$_{25D}$ values rescaled by 10^4 (basis points) for readability; multiply coefficients by 10^{-4} to recover raw fraction units.

Table 19: H18 CEO 3-IV Decomp: ClarityCEO + UncResCEO (DWZ Eq.5 Full) + UncPreCEO (raw) and SEC Conference-Call Comment Letter Receipt (LPM, $[call_t, call_{t+1}]$ window)

	(1)	(2)	(3)	(4)	(5)	(6)
	CCCL					
CEO Clarity (DWZ)	0.0059	0.0039	0.0062	0.0040	0.0058	0.0040
	(0.0015)	(0.0030)	(0.0015)	(0.0030)	(0.0015)	(0.0030)
CEO Residual Uncertainty (DWZ)	-0.0003	-0.0003	-0.0003	-0.0003	-0.0004	-0.0004
	(0.0009)	(0.0009)	(0.0009)	(0.0009)	(0.0009)	(0.0009)
CEO Pres Uncertainty	0.0008	**0.0014***	**0.0016****	**0.0014***	0.0007	**0.0013***
	(0.0008)	(0.0009)	(0.0007)	(0.0009)	(0.0008)	(0.0009)
lnAssets	**-0.0006****	**0.0018***	0.0001	0.0017	**-0.0006****	**0.0019***
	(0.0002)	(0.0011)	(0.0001)	(0.0010)	(0.0003)	(0.0011)
TobinsQ	-0.0003	0.0002	-0.0001	0.0002	-0.0002	0.0004
	(0.0003)	(0.0004)	(0.0003)	(0.0004)	(0.0003)	(0.0004)
ROA	0.0006	0.0012	-0.0014	-0.0006	-0.0006	-0.0004
	(0.0034)	(0.0052)	(0.0052)	(0.0062)	(0.0051)	(0.0062)
Leverage	-0.0008	-0.0019	-0.0009	-0.0011	-0.0009	-0.0012
	(0.0015)	(0.0034)	(0.0015)	(0.0035)	(0.0015)	(0.0035)
Capex	-0.0029	0.0106	0.0004	0.0111	-0.0033	0.0107
	(0.0063)	(0.0104)	(0.0064)	(0.0104)	(0.0066)	(0.0103)
CashRatio	-0.0014	**-0.0075***	0.0009	**-0.0080***	-0.0014	**-0.0082***
	(0.0027)	(0.0044)	(0.0026)	(0.0044)	(0.0028)	(0.0044)
DivDummy	-0.0005	0.0003	-0.0005	0.0002	-0.0005	0.0002
	(0.0008)	(0.0015)	(0.0008)	(0.0015)	(0.0008)	(0.0015)
sCFO	-0.0007	-0.0020	-0.0004	-0.0019	-0.0007	-0.0019
	(0.0005)	(0.0024)	(0.0004)	(0.0024)	(0.0005)	(0.0024)
Lagged_DV	**-0.0079*****	**-0.0409*****	**-0.0077*****	**-0.0410*****	**-0.0074*****	**-0.0405*****
	(0.0008)	(0.0027)	(0.0007)	(0.0027)	(0.0008)	(0.0028)
SalesGrowth			-0.0002	-0.0013	-0.0001	-0.0011
			(0.0008)	(0.0011)	(0.0008)	(0.0011)
RDSales			-0.0003	-0.0002	-0.0003	-0.0001
			(0.0002)	(0.0002)	(0.0002)	(0.0002)
CashFlowAt			0.0025	0.0043	0.0009	0.0037
			(0.0050)	(0.0064)	(0.0050)	(0.0064)
DailyVola			0.0000	**-0.0000***	0.0000	-0.0000
			(0.0000)	(0.0000)	(0.0000)	(0.0000)
Extended Controls			Yes	Yes	Yes	Yes
Industry FE	Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes
Year FE	Yes	Yes	Yes	Yes		
Year-Quarter FE					Yes	Yes
N	44,113	44,113	44,096	44,096	44,096	44,096
R^2	0.001	0.002	0.000	0.002	0.001	0.002
Adj. R^2	-0.000	-0.032	-0.001	-0.032	-0.001	-0.033

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). DWZ Full decomposition: Clarity = -CEO FE; UncRes = call-level residual. R^2 includes absorbed fixed effects (not within-R^2).

Table 20: Phase E CEO sudden-death DiD: ΔCash around exogenous CEO shock; PSM 1:1 nearest-neighbor controls on Bates (2009) covariates at $t-1$

FE	(1) Industry Ghafoor (2023)	(2) Firm	(3) Ind+YQ	(4) Firm+YQ
Treated	+0.0352***		+0.0326***	
	(0.0129)		(0.0123)	
Post	+0.0212**	+0.0033	+0.0384***	+0.0264*
	(0.0094)	(0.0086)	(0.0096)	(0.0140)
Treated × Post (ATT)	-0.0178	-0.0090	-0.0162	-0.0053
	(0.0163)	(0.0141)	(0.0138)	(0.0127)
N (firm-quarters)	338	338	338	338
R^2	0.406	0.320	0.420	0.320
N (firms)	16	16	16	16

8 viable treated firms (CEO sudden death 2002-2018; ±12 quarter coverage; ≥5 pre-event UncAnsCEO obs) matched 1:1 to non-event controls via propensity-score nearest-neighbor on lnAssets, Leverage, ROA, DivDummy, sCFO at $t-1$. Window: ±12 fiscal quarters around death event. Col (1) replicates Ghafoor (2023) §4.5 specification (industry FE; firm-clustered SE). Standard errors clustered at firm level; with 16 clusters (8 treated + 8 PSM-matched controls) asymptotic cluster-robust inference may be downward-biased; results interpreted as suggestive (advisor-recommended). Parallel-trends placebo (Treated × Pre-4q): β=+0.0266, p=0.183; Treated × Pre-8q: β=+0.0210, p=0.146. Both insignificant supports DiD identification. PSM balance (std. diff): lnAssets=+0.04, Leverage=-0.08, ROA=-0.04, DivDummy=+0.00, sCFO=+0.19 (all | < 0.25|, Rosenbaum-Rubin standard). Coefficients on Lagged DV and other controls suppressed for space. ATT p-values are one-tailed (Ghafoor 2023 negative prior). $^{*}/^{**}/^{***}$ denote $p < 0.10/0.05/0.01$.

Table 21: DWZ §6 First-Difference Replication: CEO Turnover and Cash Holdings

FE	(1) DWZ §6 FF12	(2) + Bates FF12	(3) Ind-adj demeaned
Δ ClarityCEO	-0.0183**	-0.0172*	-0.0182**
	(0.0109)	(0.0107)	(0.0108)
Δ NegCall	-0.0390**	-0.0369**	-0.0387**
	(0.0162)	(0.0163)	(0.0161)
Δ UncPreCEO	-0.0002	-0.0053	-0.0002
	(0.0156)	(0.0151)	(0.0156)
Δ UncQue	+0.0221	+0.0194	+0.0223
	(0.0157)	(0.0156)	(0.0156)
Δ lnAssets	-0.0321***	-0.0280***	-0.0322***
	(0.0072)	(0.0074)	(0.0072)
Δ ROA	+0.0645	+0.0532	+0.0653
	(0.0592)	(0.0578)	(0.0591)
Δ Leverage		-0.0686**	
		(0.0321)	
Δ DivDummy		+0.0118	
		(0.0129)	
Δ sCFO		+0.2054***	
		(0.0766)	
N (turnover pairs)	659	659	659
R^2	0.108	0.128	0.059

Sample: 661 CEO turnover pairs in F1D Main-industry firms 2002Q1–2018Q4. One observation per turnover pair; variables averaged over each CEO's tenure at the firm. WLS weighted by harmonic-mean of quarter counts. Replicates Dzielinski, Wagner & Zeckhauser (2021) §6 Equation 7 first-difference design, adapted from Tobin's Q to cash holdings. One observation per CEO turnover pair (gvkey, old, new); variables averaged over each CEO's tenure at the firm; Δ = New – Old. Estimated by weighted least squares with weights $w = (n_{\text{old}} \cdot n_{\text{new}})/(n_{\text{old}} + n_{\text{new}})$ per DWZ verbatim. Robust (HC1) standard errors. Sample filters per DWZ §6 verbatim: gap between old CEO last call and new CEO first call ≤ 120 days; both CEOs ≥ 5 calls at firm; both CEOs have valid ClarityCEO. Deviations from DWZ verbatim: FF12 industry FE (vs FF48); intangibles ratio omitted (not in panel); sample = F1D Main industries (FF12 ∉ {Finance, Utility}). **Endogeneity caveat** (§4.3 disclosure): DWZ themselves note this design "does not completely eliminate endogeneity concerns" (§6); because cash is a CEO-discretion choice variable, the endogeneity profile here is more concerning than in DWZ's Tobin's Q application. This table is a robustness companion to the Phase E sudden-death DiD (Table 20); forced/voluntary turnover not separated. p-value on Δ ClarityCEO is one-tailed (H1 prior: clarity ↑⇒ cash ↓); other p-values two-tailed. $^{*}/^{**}/^{***}$ denote $p < 0.10/0.05/0.01$.

Table 22: Lewbel (2012) Heteroskedasticity-Based IV: Cash Holdings

	(1) OLS	(2) 2SLS-Lewbel	(3) + extended
UncResCEO	+0.0021**	+0.0100	+0.0101
	(0.0010)	(0.0083)	(0.0085)
N	43,471	43,471	43,454
N firms	1,423	1,423	1,423
R^2	0.863	0.863	0.864
First-stage F	n/a	20.4	41.3
Sargan p	n/a	0.919	0.009
Wu-Hausman p	n/a	0.241	0.159

Sample: HC Main panel — F1D firm-quarters in Main industries (FF12 ∉ {Finance, Utility}) 2002Q1–2018Q4. Identical to H1.ceo2 main-panel HC sample. Lewbel (2012) heteroskedasticity-based IV identifies β on UncResCEO via internal instruments $Z_j = (W_j - \bar{W}_j) \cdot \hat{\varepsilon}_X^{\text{1st-stage}}$ for each W_j with significant Pesaran-Taylor heteroskedasticity in the first-stage residual. Col (1) OLS baseline replicates the H1.ceo2 main panel HC specification on the same sample. Cols (2)-(3) are 2SLS with Lewbel-manufactured instruments. Pesaran-Taylor p-values ($W_j \rightarrow$ heteroskedasticity in $\hat{\varepsilon}_X$): Leverage=0.000, lnAssets=0.000, TobinsQ=0.000, ROA=0.004, Capex=0.000, DivDummy=0.733, sCFO=0.119, CashRatio_lag=0.017. Standard errors firm-clustered. **Identification scope (§4.3 disclosure):** Lewbel addresses omitted TIME-VARYING confounders within firm spells (which neither firm-FE nor manager-FE absorbs). For reverse-causality concerns (firms hoard cash → CEO talks more uncertainly about that position), the design is weaker than Phase E sudden-death (Table 20); the three designs together cover three different identification threats rather than converge on a single coefficient. p-value on UncResCEO is one-tailed (H1 prior: UncResCEO ↑⇒ Cash ↑). $^{*}/^{**}/^{***}$ denote $p < 0.10/0.05/0.01$.